PROVEIT / RESEARCH CONTINUATION

Differential Polygamma and Lerch Identities

Exact residue cancellation, higher Gamma moments, spectral derivatives, and harmonic-tail reconstruction

Prepared with ChatGPT for Vladimir Reshetnikov and the ProveIt project
11 October 2026

Abstract

We continue six newly submitted reports and the current ProveIt polylogarithm manuscript. The main theorem extends the normalized digamma-power calculus to every differential polynomial in the digamma function. Its asymptotic residue and its large-circle constant depend only on the scalar digamma part. In particular, every monomial containing a positive-order polygamma has an exact finite-part integral expressed by convergent multiple-zeta values and convergent spectral derivatives, with no singular Laurent coordinate. A triangular integration-by-parts theorem gives a canonical formal basis and retains every endpoint constant. We give normalized primitives, a common-shift endpoint law, polylogarithmic kernels, and an explicit formula for every bilinear polygamma integral.

The Gamma–Appell continuation isolates the highest height-one Laurent coordinate at every degree and makes the remaining coordinates convergent; the sixth and seventh degrees are worked out. A second continuation evaluates all spectral jets of the centered continuous-depth Lerch family on its odd lattice, including mixed depth derivatives and Stieltjes specializations. Finally, an exact determinant formula proves the optimal-prefix conjecture for every neighboring pair of even harmonic-tail orders. The unrestricted unequal-order conjecture and the Gaussian S_6, S_8 candidates remain open. Proofs, source comparisons, and independent symbolic and numerical checks are supplied; numerical agreement is never used as a proof of an identity.

Source baseline: ProveIt revision `f7e35be357654198c1546fd40f363694c5bcc2cc`.
The contribution is measured against the inspected sources. A claim of historical priority for every displayed specialization is not made.

Contents

1 Research targets and conventions **1**
1.1 Functions, finite parts, and spectral conventions 1

2 A differential-polynomial finite-part calculus **2**
2.1 Finite local data and the global normalization 2
2.2 The scalar projection controls both boundary obstructions 3
2.3 The complete spectral formula 4

3 Convergent zeta coordinates and explicit product identities **5**
3.1 A finite reduction without singular word coordinates 5
3.2 Polylogarithmic kernels for the same coordinates 6
3.3 Every bilinear polygamma integral 6

4 Canonical integration by parts with endpoint constants **7**
4.1 The endpoint map is part of the theorem 8

5 Normalized primitives and common shifted endpoints **9**
5.1 A multiple-polylogarithm representation of the primitives 10

6 One singular Laurent coordinate at every Gamma–Appell degree **10**
6.1 The normalized moments and their finite data 11
6.2 Why the correction is a finite convergent derivative family 12
6.3 Explicit sixth and seventh normalized powers 14
6.4 Recovering raw moments without external tables 15
6.5 Proof ledger for the two higher-degree identities 16
6.6 Validation and the remaining arithmetic question 17

7 Every order derivative on the odd Lerch lattice **17**
7.1 Functions, branches, and normalization 18
7.2 The finite formula for all mixed jets 19
7.3 A convergent elementary series for every reflected jet 20
7.4 Independent Gamma-moment realization 21
7.5 Stieltjes coordinates at depth two 21
7.6 Gamma-weighted polylogarithm order derivatives 22
7.7 What this resolves and what remains 24

8 Optimal initial moments for adjacent even tail orders **24**
8.1 The inherited coordinates and the exact target 24
8.2 A positive recurrence and the determinant 25
8.3 Proof by rotation of the local differential equation 26
8.4 Explicit reconstruction without matrix inversion 28
8.5 An infinite family of ordinary rational evaluations 28
8.6 A convergent rational polygamma identity at orders four and six 29

8.7 A smaller boundary determinant for a fixed gap 30

9 Source audit, proof status, and further research 31

9.1 What has changed relative to the source questions 31
9.2 Corrections and editorial recommendations 32
9.3 Verification and the role of the accompanying artifacts 33
9.4 Further research questions 33
9.4.1 The boundary determinant for arbitrary even gaps 33
9.4.2 Systematic rational telescoping for tail products 34
9.4.3 The spectral image of the canonical differential basis 34
9.4.4 Relations among the sixth and seventh mixed directions 34
9.4.5 Higher spectral derivatives of the residue calculus 34
9.4.6 Rational shifts and multiple-Hurwitz coordinates 34
9.4.7 Additional reductions at rational noninteger depth 34
9.4.8 Independent twists and noncentered Gamma weights 35
9.4.9 Four or more harmonic tails 35
9.4.10 Functional certificates for the remaining ladders 35
9.4.11 Formal certificates with a small analytic interface 35

References 35

1 Research targets and conventions

The newest incoming reports make the next exact questions unusually specific. The higher-digamma report asks for a calculus of arbitrary differential polynomials, for common shifted endpoints, and for explicit sixth and seventh moments [2]. The canonical fifth-moment report asks which differential monomials survive integration by parts, with the finite endpoint value retained, and conjectures that an optimal initial set of harmonic-tail moments has full formal rank [3]. The continuous-depth report asks for spectral derivatives at its odd evaluation lattice [5]. These are the principal targets here.

The normalized Mittag–Leffler expansion for scalar powers, the height-one Gamma integral, the centered Lerch reflection, and the original two-tail moment formulas are antecedents. We reproduce the parts needed to fix conventions and then prove their extensions. This distinction is important because the repository already contains several independently derived versions of related calculations [1, 4, 6, 7]. The present audit is targeted, not an audit of every analytic assertion in the collection.

Source question	Result and remaining scope
Differential-polynomial moments	A complete residue and finite-part theorem, with a finite reduction to convergent zeta coordinates for every mixed monomial.
Integration by parts at all weights	An explicit direct-sum normal form; its dimension is a formal algebraic dimension, not independence of the resulting numbers.
Higher scalar Gamma normalization	Only the highest height-one Laurent coordinate is singular; all remaining coordinates have convergent representatives. Arithmetic reduction of those coordinates remains a separate question.
Odd-lattice Lerch spectral jets	Every order and mixed depth derivative, with explicit local Gamma coefficients and convergent spectral evaluators.
Optimal initial tail moments	Proved for all neighboring even orders $p = q - 2$. Arbitrary even gaps remain open.

1.1 Functions, finite parts, and spectral conventions

Write $\psi = \Gamma'/\Gamma$ and $\psi^{(r)} = \mathrm{d}^r\psi/\,\mathrm{d}x^r$, with $\psi^{(0)} = \psi$. Let

$$H_n^{(r)} = \sum_{k=1}^{n} k^{-r}, \qquad H_n = H_n^{(1)}, \qquad A_r(n) = \zeta(r) - H_n^{(r)} \quad (r \geq 2).$$

The empty sum is zero. A strict word has the convention

$$H_n(w_1, \ldots, w_d) = \sum_{n \geq k_1 > \cdots > k_d \geq 1} \prod_{j=1}^{d} k_j^{-w_j}, \qquad H_n(\varnothing) = 1.$$

For convergent arguments, $Z(s_1, \ldots, s_d)$ is the corresponding infinite strict multiple-zeta series. A derivative $\partial_j Z$ means differentiation in its jth spectral argument; its summand contains $-\log k_j$. Classical definitions and normalizations of Gamma, Hurwitz zeta, and Lerch's transcendent are those of the DLMF [8, 9, 10].

The Stieltjes convention is

$$\zeta(1+s, a) = \frac{1}{s} + \sum_{j \geq 0} \frac{(-1)^j}{j!} \gamma_j(a) s^j, \qquad \operatorname{Re} a > 0. \tag{1.1}$$

We write $\gamma_j = \gamma_j(1)$ and $\gamma = \gamma_0$. The notation $[s]F(s)$ denotes a Laurent coefficient. It agrees with $F'(0)$ only when F is holomorphic at zero.

Suppose $f(x) = \sum_{j=1}^{M} p_j x^{-j} + O(1)$ near 0. Our finite part is

$$\mathrm{FP}\int_0^1 f(x)\,\mathrm{d}x = \lim_{\varepsilon\downarrow 0}\left\{\int_\varepsilon^1 f(x)\,\mathrm{d}x - \sum_{j=2}^{M}\frac{p_j\varepsilon^{1-j}}{j-1} + p_1\log\varepsilon\right\}. \tag{1.2}$$

The coordinate is exactly x and the upper endpoint exactly 1. Every finite-part statement below uses this definition. Normalized primitives use principal logarithms on $\mathbb{C}\setminus(-\infty,0]$.

For a polynomial $P \in \mathbb{C}[u_0, u_1, \ldots, u_R]$, set

$$f_P(x) = P(\psi(x), \psi'(x), \ldots, \psi^{(R)}(x)), \qquad \phi_P(Y) = P(Y, 0, \ldots, 0),$$
$$\mathcal{I}(P) = \mathrm{FP}\int_0^1 f_P(x)\,\mathrm{d}x.$$

We call P mixed when $\phi_P = 0$. This is a precise algebraic condition: each of its monomials contains at least one u_r with $r \geq 1$. The word does not assert that the associated numerical periods are independent. Give u_r differential weight $r+1$.

2 A differential-polynomial finite-part calculus

2.1 Finite local data and the global normalization

For $n \geq 0$, put $b_n = H_n - \gamma$ and $c_{r,n} = H_n^{(r)} + (-1)^r\zeta(r)$ for $r \geq 2$. The familiar Gamma recurrence and Taylor expansion give

$$v_n(\varepsilon) := \psi(-n+\varepsilon) = -\varepsilon^{-1} + b_n + \sum_{r\geq 2} c_{r,n}\varepsilon^{r-1}. \tag{2.1}$$

This series converges for $0 < |\varepsilon| < 1$; its differentiated series give all polygammas. Let M be an upper bound for the differential weights of the monomials of P. Define

$$p_j(n) = [\varepsilon^{-j}]P(v_n, v_n', \ldots, v_n^{(R)}), \quad P_n(z) = \sum_{j=1}^{M}\frac{p_j(n)}{(z+n)^j}, \quad \kappa_P = [\varepsilon^0]f_P(\varepsilon). \tag{2.2}$$

All coefficients are finite polynomials in the stated harmonic numbers and zeta values. To find the coefficients through the constant term, one may truncate v_n after ε^{M-1} before differentiating. For a homogeneous monomial of weight M, the regular coefficient of harmonic order r raises the exponent relative to its all-pole term by r; hence orders above M cannot contribute. Linearity handles inhomogeneous polynomials.

Theorem 2.1 (Normalized differential-polynomial expansion)**.** *Away from the nonpositive integers,*

$$f_P(z) = P_0(z) + \kappa_P + \sum_{n\geq 1}\bigl(P_n(z) - P_n(0)\bigr). \tag{2.3}$$

The series converges normally on compact subsets of its domain. The entire remainder is exactly the displayed constant.

Proof. Each $p_j(n)$ grows at most as a fixed power of $1+\log(n+1)$: only H_n is unbounded, whereas the higher harmonic numbers are bounded. On a fixed compact set, $(n+z)^{-j} - n^{-j} = O(n^{-j-1})$. This proves normal convergence. To determine the remainder, put $g(w) = f_P(w) - P_0(w)$ and integrate $zg(w)/(w(w-z))$ on $|w| = N + 1/2$.

The right-half-plane asymptotics give $\psi(w) = O(1+\log N)$ and bounded fixed positive derivatives. On the left use $\psi(w) = \psi(1-w) - \pi\cot(\pi w)$ and its derivatives. The cotangent and

each fixed derivative are uniformly bounded on these circles. For $|\operatorname{Im} w| \geq 1$ this follows from their exponential representations; for $|\operatorname{Im} w| < 1$ the circles remain a fixed positive distance from the integers. Thus $g = O((1+\log N)^d)$ for some fixed d, and the contour integral tends to zero. The residues at z and 0 are $g(z)$ and $-\kappa_P$; a pole at $-n$ contributes $-P_n(z) + P_n(0)$. The residue theorem gives the assertion. □

2.2 The scalar projection controls both boundary obstructions

Define polynomials in L by

$$C_P(L) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \phi_P(L + \mathrm{i}\theta)\,\mathrm{d}\theta, \qquad C_P = C_P(0), \tag{2.4}$$

$$R_P(L) = -\frac{\phi_P(L+\mathrm{i}\pi) - \phi_P(L-\mathrm{i}\pi)}{2\mathrm{i}\pi}. \tag{2.5}$$

The integrals are integrals of polynomials, so these are finite algebraic formulas. If $\phi_P(Y) = \sum a_k Y^k$, then

$$C_P = \sum_{m\geq 0} a_{2m}\frac{(-1)^m \pi^{2m}}{2m+1}. \tag{2.6}$$

Theorem 2.2 (Scalar projection and residue cancellation)**.** *For every fixed P, with $r_P(n) = p_1(n)$, there is an integer d such that*

$$\kappa_P - \sum_{n=1}^{N} P_n(0) = C_P(\log N) + o(1), \tag{2.7}$$

$$r_P(n) = R_P(\log n) + O((1+\log n)^d/n), \tag{2.8}$$

$$r_P(n) - r_P(n-1) = \frac{R_P'(\log n)}{n} + O((1+\log n)^d/n^2). \tag{2.9}$$

In particular, a mixed polynomial has $C_P = R_P = 0$.

Proof. The first identity follows by integrating $f_P(w)/w$ on the same circles. Away from an arc of angular length $O(\log N/N)$ around the negative real axis, reflection and the standard sectorial asymptotics give $\psi(w) = \operatorname{Log} w + O(1/N)$ and $\psi^{(r)}(w) = O(N^{-r}) + O(N^{-2\pi})$ for fixed $r \geq 1$. On the exceptional arc all fixed polygammas are bounded and the digamma is $O(\log N)$. Every mixed monomial consequently has angular average $o(1)$. The scalar terms have the stated logarithmic average. Replacing $\log(N+1/2)$ by $\log N$ changes the answer by $o(1)$.

For the residue, take the constant-tail specialization of the local series: b_n tends formally to L, and $c_{r,n}$ tends to $(1+(-1)^r)\zeta(r)$. The resulting series is $L + y$, where $y = -\pi\cot(\pi\varepsilon)$. Its derivatives satisfy

$$y' = y^2 + \pi^2, \qquad y^{(r)} = Q_r(y), \quad Q_1(Y) = Y^2 + \pi^2, \quad Q_{r+1}(Y) = (Y^2+\pi^2)Q_r'(Y).$$

Every Q_r with $r \geq 1$ is divisible by $Y^2 + \pi^2$. Under the local change of variable $Y = y$, residue extraction is

$$\operatorname*{Res}_{\varepsilon=0} P(L+y, y', \ldots) = -[Y^{-1}]\frac{P(L+Y, Q_1(Y), Q_2(Y), \ldots)}{Y^2+\pi^2}, \tag{2.10}$$

where the coefficient is in the Laurent expansion at infinity. The minus sign follows from $y \sim -1/\varepsilon$. Every mixed term in the quotient is a polynomial and contributes zero. For the scalar term, polynomial division leaves a remainder $AY + B$, where $A = (\phi_P(L+\mathrm{i}\pi) - \phi_P(L-\mathrm{i}\pi))/(2\mathrm{i}\pi)$. This proves the exact residue formula (2.5).

Finally, the harmonic asymptotic expansions through their $1/n$ terms give $r_P(n) = R_P(\log n) + n^{-1}U(\log n) + O((1+\log n)^d/n^2)$ for a polynomial U. Differencing these explicit terms gives (2.9). An unspecified $O(1/n)$ remainder is not being differentiated or differenced to claim a stronger error estimate. □

2.3 The complete spectral formula

Write $\Delta p_j(n) = p_j(n) - p_j(n-1)$ and set

$$J_P = \sum_{n\geq 1}\sum_{j=2}^{M} \frac{\Delta p_j(n)}{(j-1)n^{j-1}}, \qquad D_P(s) = \sum_{n\geq 1} \frac{\Delta p_1(n)}{n^s} \quad (\operatorname{Re} s > 0). \tag{2.11}$$

The first sum is absolutely convergent. To see this directly, replace $H_n^{(r)}$ by $H_{n-1}^{(r)} + n^{-r}$ in the finite pole polynomial; each difference gains at least one factor $1/n$.

Theorem 2.3 (Every differential-polynomial finite part)**.** *The Dirichlet series D_P continues meromorphically to* $\operatorname{Re} s > -1$*, and*

$$\boxed{\mathcal{I}(P) = C_P + J_P + [s]D_P(s).} \tag{2.12}$$

If $R_P'(L) = \sum a_j L^j$, an entirely ordinary-series form is

$$\mathcal{I}(P) = C_P + J_P - \sum_j a_j\gamma_{j+1} - \sum_{n\geq 1}\left\{\Delta p_1(n) - \frac{R_P'(\log n)}{n}\right\}\log n. \tag{2.13}$$

The last series converges absolutely. If ϕ_P has degree $d \geq 2$ and leading coefficient a, then D_P has a pole of exact order $d-1$ at zero, with leading term $-a\,d!s^{1-d}$. If $\deg\phi_P \leq 1$, including $\phi_P = 0$, it is holomorphic there.

Proof. The asymptotic in (2.9) gives the normal decomposition

$$D_P(s) = \sum_j a_j(-1)^j\zeta^{(j)}(1+s) + \sum_{n\geq 1}\left\{\Delta p_1(n) - R_P'(\log n)/n\right\} n^{-s}.$$

Its second term is holomorphic on $\operatorname{Re} s > -1$, including all derivatives on compact subsets. By (1.1), its linear coefficient is exactly the Stieltjes term and the last sum in (2.13). The leading term of R_P' is $-ad(d-1)L^{d-2}$, proving the pole classification.

It remains to establish the finite-part normalization. Integrate (2.3), first through a finite cutoff N. For $j \geq 2$, finite summation by parts gives

$$\sum_{n=1}^{N} p_j(n)(n^{1-j} - (n+1)^{1-j}) = p_j(0) + \sum_{n=1}^{N}\frac{\Delta p_j(n)}{n^{j-1}} - \frac{p_j(N)}{(N+1)^{j-1}}.$$

The first term cancels exactly the finite part of the pole at zero; the last tends to zero. For simple poles the finite identity is

$$\sum_{n=1}^{N} p_1(n)\log(1+1/n) = p_1(N)\log(N+1) - \sum_{n=1}^{N}\Delta p_1(n)\log n.$$

The boundary $p_1(N)\log(N+1)$ has logarithmic polynomial $LR_P(L)$, with zero constant coefficient. The large-circle term has constant C_P. The Stieltjes limit $\sum_{n\leq N}\log^j(n)/n = \log^{j+1}N/(j+1) + \gamma_j + o(1)$ shows that the constant term of the final logarithmic sum is $-[s]D_P(s)$. The subtracted series in (2.3) is uniformly convergent after the origin pole is removed, including on $[0,1]$, so this cutoff calculation computes the finite part (1.2). This proves both formulas. □

For a mixed P the result simplifies to the ordinary convergent identity

$$\boxed{\mathcal{I}(P) = J_P + D_P'(0) = J_P - \sum_{n\geq 1}\Delta p_1(n)\log n.} \tag{2.14}$$

The disappearance of singular Laurent coordinates is thus a proved consequence of the local cotangent algebra, not a fitted cancellation.

3 Convergent zeta coordinates and explicit product identities

3.1 A finite reduction without singular word coordinates

Absolute convergence of a combined Dirichlet series does not by itself make every term in an arbitrary quasi-shuffle expansion convergent. The following construction supplies convergent coordinates individually.

Theorem 3.1 (Convergent word reduction for every mixed monomial)**.** *Let P be a mixed monomial with K factors, counting multiplicities. Then $\mathcal{I}(P)$ is a finite linear combination of convergent strict multiple-zeta values and their first spectral derivatives, of depth at most $K-1$. The coefficients belong to the ring generated over $\mathbb{Q}$ by γ and the ordinary zeta values occurring in the pole polynomials. For $K=1$ the integral is zero. Linearity gives a finite algorithm for every mixed polynomial.*

Proof. First expand the higher-pole differences in $h_r = H_{n-1}^{(r)}$ and $v = 1/n$. A coefficient $p_j(n)$ uses at least one singular factor, so its harmonic degree is at most $K-1$. Taking a difference replaces at least one regular factor by a power of v. Hence $\Delta p_j(n)$ has harmonic degree at most $K-2$, and every term contains v^q with $q \geq 1$. Finite stuffle multiplication replaces products of the h_r by strict words. Therefore

$$\sum_{n\geq 1} \frac{v^q H_{n-1}(w)}{n^{j-1}} = Z(q+j-1, w), \qquad j \geq 2,$$

with first entry at least two and depth at most $K-1$. This gives the claimed representation of J_P.

For the residue, express $p_1(n)$ as a polynomial $F(B, A_2, A_3, \ldots)$ with $B = H_n - \gamma$ and $A_r = A_r(n)$. In (2.1) this amounts to replacing $c_{r,n}$ by $(1+(-1)^r)\zeta(r) - A_r$. The residue cancellation theorem, with B an indeterminate, proves the polynomial identity $F(B, 0, 0, \ldots) = 0$. Thus every monomial of F contains at least one upper tail.

Put $b = H_{n-1} - \gamma$ and use the exact increment formula

$$\Delta p_1(n) = F(b+v, A_2, A_3, \ldots) - F(b, A_2 + v^2, A_3 + v^3, \ldots). \tag{3.1}$$

Each resulting term is a constant times

$$n^{-q} H_{n-1}(w)\, U_n(v_1, \ldots, v_d), \qquad U_n(v_1, \ldots, v_d) = \sum_{k_1 > \cdots > k_d > n} \prod k_j^{-v_j}. \tag{3.2}$$

Here all upper letters v_j are at least two. Stuffle multiplication has been applied separately above and below the distinguished index n. If no upper letter remains, at least one removed tail supplies n^{-r} with $r \geq 2$, so $q \geq 2$. If an upper letter remains, $q \geq 1$ is sufficient. The corresponding sums and logarithmic sums are

$$\sum_{n\geq 1} n^{-q} H_{n-1}(w) U_n(v) = Z(v_1, \ldots, v_d, q, w), \tag{3.3}$$

$$-\sum_{n\geq 1} n^{-q} H_{n-1}(w) U_n(v) \log n = \partial_{d+1} Z(v_1, \ldots, v_d, q, w). \tag{3.4}$$

Every displayed word has first entry at least two. Its derivative is absolutely convergent as well: a logarithm is dominated by an arbitrarily small extra power of the largest index, while the standard admissible series has a positive convergence margin. The difference in (3.1) removes at least one of at most $K-1$ harmonic or tail factors; reinserting the distinguished index gives depth at most $K-1$. Formula (2.14) completes the proof. For one positive-order polygamma all its principal coefficients are constant in n and its simple residue is zero, giving $J_P = D_P = 0$. □

This theorem makes a finite-coordinate statement. It does not assert that its resulting multiple-zeta derivatives reduce to ordinary zeta derivatives, nor that distinct remaining coordinates are independent. For polynomial coefficients already containing other constants, those constants are simply retained as coefficients.

3.2 Polylogarithmic kernels for the same coordinates

Let

$$S_j(z) = \sum_{n\geq 1} \Delta p_j(n) z^n, \qquad |z| < 1. \tag{3.5}$$

The finite harmonic expansion shows that every S_j is a finite linear combination of strict multiple polylogarithms:

$$\sum_{n\geq 1} \frac{H_{n-1}(w)}{n^q} z^n = \mathrm{Li}_{q,w}(z), \qquad \mathrm{Li}_{s_1,\dots,s_d}(z) = \sum_{n_1>\cdots>n_d\geq 1} \frac{z^{n_1}}{n_1^{s_1}\cdots n_d^{s_d}}.$$

For a mixed polynomial, S_1 has the continuous endpoint value $S_1(1) = \sum \Delta p_1(n) = -p_1(0)$.

Corollary 3.2 (Ordinary logarithmic polylogarithm bridge)**.** *For mixed P,*

$$\boxed{\mathcal{I}(P) = J_P + \int_0^1 \frac{S_1(z) - zS_1(1)}{z(-\log z)}\,\mathrm{d}z.} \tag{3.6}$$

The integral is ordinary and absolutely convergent. Moreover,

$$J_P = \sum_{j=2}^{M} \frac{1}{(j-1)!} \int_0^1 \frac{(-\log z)^{j-2}}{z} S_j(z)\,\mathrm{d}z. \tag{3.7}$$

Proof. The estimate $\Delta p_1(n) = O(n^{-2}(1+\log n)^d)$ implies $\sum |\Delta p_1(n)| \log n < \infty$. Frullani's identity in the form

$$\int_0^1 \frac{z^{n-1}-1}{-\log z}\,\mathrm{d}z = -\log n$$

therefore justifies termwise integration of the complete subtracted kernel, including absolute integrability. This gives $D_P'(0)$. For the second formula use $\int_0^1 z^{n-1}(-\log z)^{j-2}\,\mathrm{d}z = (j-2)!/n^{j-1}$ and the absolute convergence of the harmonic differences. Individual polylogarithms in a representation of S_1 can diverge at one; $S_1(1)$ means the value of their complete combination, not a termwise substitution into a divergent expression. □

3.3 Every bilinear polygamma integral

The entire bilinear layer collapses to two ordinary zeta coordinates. This is also a useful explicit check on the signs and endpoint normalization in the general theorem.

Theorem 3.3 (Bilinear evaluation)**.** *For nonnegative integers r, s with $r+s \geq 1$, put $h = r+s$, $\alpha = (-1)^{r+1}$, and $\beta = (-1)^{s+1}$. Then*

$$\boxed{\mathrm{FP}\int_0^1 \psi^{(r)}(x)\psi^{(s)}(x)\,\mathrm{d}x = h!\left\{(\alpha+\beta)\zeta'(h+1) + \left[\alpha(H_h - H_s) + \beta(H_h - H_r)\right]\zeta(h+1)\right\}.} \tag{3.8}$$

In particular, for $r \geq 1$,

$$\mathrm{FP}\int_0^1 \left(\psi^{(r)}(x)\right)^2 \mathrm{d}x = 2(-1)^{r+1}(2r)!\{\zeta'(2r+1) + (H_{2r} - H_r)\zeta(2r+1)\}. \tag{3.9}$$

Proof. Assume first $r, s \geq 1$. Multiplication of the two differentiated series in (2.1) gives

$$p_1(n) = (\alpha + \beta)h! c_{h+1,n}.$$

The parity of h shows that either $\alpha+\beta = 0$ or $h+1$ is odd; in the latter case $c_{h+1,n} = -A_{h+1}(n)$. Thus $D_P(s_0) = (\alpha + \beta)h!\zeta(h + 1 + s_0)$. The higher-pole differences give

$$J_P = \left\{ \alpha r! \sum_{k=1}^{r} \frac{(h-k)!}{(r-k)!k} + \beta s! \sum_{k=1}^{s} \frac{(h-k)!}{(s-k)!k} \right\} \zeta(h+1).$$

The finite binomial identity

$$r! \sum_{k=1}^{r} \frac{(r+s-k)!}{(r-k)!k} = (r+s)!(H_{r+s} - H_s) \tag{3.10}$$

follows, for example, by taking the logarithmic derivative in a of Vandermonde's identity for $[x^r](1-x)^{-a}(1-x)^{-s-1}$ at $a = 0$, or by induction in r. To see the coefficient calculation directly, $\partial_a (a)_k/k!|_{a=0} = 1/k$ for $k \geq 1$, and the derivative of $(a+s+1)_r/r!$ is $\binom{r+s}{r}(H_{r+s} - H_s)$. Multiplication by $r!s!$ gives (3.10).

If $r = 0 < s$, the coefficient of the pole of order $s + 1$ contains $\beta s! b_n$ instead. Its difference is $\beta s!/n$, and it supplies exactly the last $k = s$ term of the same finite sum. The simple-pole formula remains valid. The case $s = 0 < r$ is symmetric. Formula (2.14) now gives the result. □

For example,

$$\mathrm{FP} \int_0^1 (\psi')^2 \, \mathrm{d}x = 2\zeta(3) + 4\zeta'(3), \tag{3.11}$$

$$\mathrm{FP} \int_0^1 \psi\psi'' \, \mathrm{d}x = -3\zeta(3) - 4\zeta'(3), \tag{3.12}$$

$$\mathrm{FP} \int_0^1 \psi'\psi'' \, \mathrm{d}x = -3\zeta(4), \tag{3.13}$$

$$\mathrm{FP} \int_0^1 (\psi'')^2 \, \mathrm{d}x = -28\zeta(5) - 48\zeta'(5), \tag{3.14}$$

$$\mathrm{FP} \int_0^1 \psi'\psi''' \, \mathrm{d}x = 32\zeta(5) + 48\zeta'(5). \tag{3.15}$$

If r and s have opposite parity, the zeta derivative disappears exactly. The exceptional pair $(0, 0)$ has a singular scalar residue and instead gives $\mathcal{I}(u_0^2) = 2\gamma_1 - 2\zeta(2)$ by Theorem 2.3.

4 Canonical integration by parts with endpoint constants

Let $\mathcal{A} = \mathbb{Q}[u_0, u_1, u_2, \ldots]$ and

$$\mathcal{D} = \sum_{r \geq 0} u_{r+1} \frac{\partial}{\partial u_r}, \qquad \mathrm{wt}(u_r) = r + 1.$$

Each polynomial uses finitely many variables. The weight-w subspace $\mathcal{A}_w$ has dimension $p(w)$, the partition number. Let $\mathcal{B}_w$ consist of u_0^w and all weight-w monomials whose highest positive-index variable occurs at least twice. For $w = 0$ use $\mathcal{B}_0 = \{1\}$.

Theorem 4.1 (A triangular normal form at every weight)**.** *For every $w \geq 2$ there is a direct sum*

$$\mathcal{A}_w = \mathcal{D}\mathcal{A}_{w-1} \oplus \operatorname{span}_{\mathbb{Q}} \mathcal{B}_w. \tag{4.1}$$

Every $P \in \mathcal{A}_w$ therefore has a unique decomposition $P = \mathcal{D}Q + N$ with $Q \in \mathcal{A}_{w-1}$ and N in the displayed normal space. In particular

$$|\mathcal{B}_w| = p(w) - p(w-1) \quad (w \geq 2). \tag{4.2}$$

The quotient has dimension one at weights zero and one, so its Hilbert series is

$$t + \prod_{j \geq 2} (1 - t^j)^{-1}. \tag{4.3}$$

Proof. Suppose a monomial has highest variable u_r, $r \geq 1$, occurring exactly once. Write it as $u_r u_{r-1}^a V$, where V involves only $u_0, \ldots, u_{r-2}$. Then

$$u_r u_{r-1}^a V = \mathcal{D}\left(\frac{u_{r-1}^{a+1} V}{a+1}\right) - \frac{u_{r-1}^{a+1} \mathcal{D}V}{a+1}. \tag{4.4}$$

Every monomial in the last term has strictly smaller highest index. Iterating therefore terminates and proves that the two spaces span.

For directness, let Q be a nonconstant polynomial with highest variable u_m. The highest-index terms of $\mathcal{D}Q$ are $u_{m+1}\partial Q/\partial u_m$, all with their highest variable occurring once. They cannot occur in a polynomial in the normal space. Since the characteristic is zero, $\partial Q/\partial u_m \neq 0$. This also proves that $\mathcal{D}$ is injective on every positive-weight subspace. Taking dimensions proves (4.2). The weight-zero kernel consists of constants, which explains the extra t in (4.3). □

Weight	Dimension	A canonical basis, with $u_r = \psi^{(r)}$
2	1	u_0^2
3	1	u_0^3
4	2	$u_0^4,\ u_1^2$
5	2	$u_0^5,\ u_0 u_1^2$
6	4	$u_0^6,\ u_0^2 u_1^2,\ u_1^3,\ u_2^2$
7	4	$u_0^7,\ u_0^3 u_1^2,\ u_0 u_1^3,\ u_0 u_2^2$
8	7	$u_0^8,\ u_0^4 u_1^2,\ u_0^2 u_1^3,\ u_1^4, u_0^2 u_2^2,\ u_1 u_2^2,\ u_3^2$

4.1 The endpoint map is part of the theorem

For a polynomial Q, define

$$E(Q) = f_Q(1) - [x^0] f_Q(x). \tag{4.5}$$

Both terms are finite polynomial calculations. The values at one are $\psi(1) = -\gamma$ and $\psi^{(r)}(1) = (-1)^{r+1} r! \zeta(r+1)$ for $r \geq 1$. The constant at zero is obtained from (2.1) with $n = 0$; it is not obtained merely by multiplying the individual constant terms.

Corollary 4.2 (Exact finite-part integration by parts)**.** *For the normal decomposition $P = \mathcal{D}Q + N$,*

$$\boxed{\mathcal{I}(P) = E(Q) + \mathcal{I}(N).} \tag{4.6}$$

For a common shift a with $\operatorname{Re} a > 0$, the corresponding ordinary identity is

$$\int_0^1 f_P(a+x)\,\mathrm{d}x = f_Q(a+1) - f_Q(a) + \int_0^1 f_N(a+x)\,\mathrm{d}x. \tag{4.7}$$

Proof. On $[\varepsilon, 1]$ the integral of $f_{\mathcal{D}Q}$ is $f_Q(1) - f_Q(\varepsilon)$. Extracting its original-coordinate finite part retains precisely $f_Q(1) - [x^0]f_Q(x)$. The shifted identity is the ordinary fundamental theorem of calculus. □

For example, $\mathcal{D}(u_0u_1) = u_1^2 + u_0u_2$, but

$$E(u_0u_1) = -\zeta(3), \qquad \mathcal{I}(u_1^2) + \mathcal{I}(u_0u_2) = -\zeta(3). \tag{4.8}$$

Likewise $E(u_1u_2) = 4\zeta(5)$. These agree with all five bilinear specializations above. Dropping total derivatives would lose these constants and give a false relation.

The dimension theorem answers the stated formal classification problem before imposing arithmetic identities among zeta coordinates. Its numerical image may have smaller dimension. For instance the u_2^2 coordinate at weight six already has the elementary zeta-derivative evaluation (3.9). Neither the formal normal form nor a failure to find further numerical relations proves arithmetic independence.

5 Normalized primitives and common shifted endpoints

The global expansion also answers the endpoint-normalization part of the common-shift question. It yields convergent formulas at independent arguments without replacing ordinary harmonic numbers by shifted ones without justification.

For $j \geq 1$ put

$$F_1(z) = \operatorname{Log} z, \qquad F_j(z) = \frac{z^{1-j}}{1-j} \quad (j \geq 2).$$

With the data of (2.2), define

$$\begin{aligned}\mathcal{P}_P(z) = &\sum_{j=1}^{M} p_j(0)(F_j(z) - F_j(1)) + \kappa_P(z-1) \\ &+ \sum_{n\geq 1}\sum_{j=1}^{M} p_j(n)\left\{F_j(n+z) - F_j(n+1) - \frac{z-1}{n^j}\right\}.\end{aligned} \tag{5.1}$$

Theorem 5.1 (Primitive and shift theorem)**.** *The function $\mathcal{P}_P$ is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, its series converges normally there, and*

$$\mathcal{P}_P'(z) = f_P(z), \qquad \mathcal{P}_P(1) = 0. \tag{5.2}$$

In particular, for $\operatorname{Re} a > 0$*,*

$$I_P(a) := \int_0^1 f_P(a+x)\,\mathrm{d}x = \mathcal{P}_P(a+1) - \mathcal{P}_P(a). \tag{5.3}$$

As $a \to 0$ in any fixed closed sector $|\arg a| \leq \pi - \delta$,

$$I_P(a) = \sum_{j=2}^{M} \frac{p_j(0)}{j-1} a^{1-j} - p_1(0)\operatorname{Log} a + \mathcal{I}(P) + O(a). \tag{5.4}$$

The last expansion refers to analytic continuation of the difference in (5.3); on the positive real axis it is an ordinary integral. Thus subtracting its displayed singular terms recovers exactly the finite part at zero.

Proof. For z in a compact set, the braces in (5.1) are $O(n^{-j-1})$ by integrating $(n+z)^{-j}-n^{-j}$ from 1 to z. The pole coefficients have only polylogarithmic growth, so the series and its derivative converge normally. The derivative is (2.3), and every term vanishes at one. This fixes the integration constant. Integrating along the horizontal segment from a to $a+1$ proves (5.3). Near zero, integrate the convergent Laurent expansion of f_P; its analytic remainder has an $O(a)$ primitive. The upper endpoint contributes another $O(a)$ term. The remaining constant is the definition (1.2). The local convergent expansion also proves uniformity on the stated sectors. □

There is an exact differential check using only the Gamma recurrence:

$$I_P'(a) = P\left(\psi(a)+a^{-1}, \psi'(a)-a^{-2}, \dots, \psi^{(r)}(a)+(-1)^r r! a^{-r-1}, \dots\right) - f_P(a). \tag{5.5}$$

This follows from $f_P(a+1)-f_P(a)$ and is valid throughout $\operatorname{Re} a > 0$. The explicit endpoint law is the part settled here. A further reduction of all such shifted logarithmic sums to a specified common-shift multiple-Hurwitz basis is not claimed.

5.1 A multiple-polylogarithm representation of the primitives

Put $V_j(z) = \sum_{n\ge 1} p_j(n) z^n$ for $|z|<1$. Finite stuffle expansion of the inclusive harmonic polynomials expresses V_j using multiple polylogarithms and rational factors $1/(1-z)$. For example, for a nonempty word w, $\sum_{n\ge 1} H_n(w) z^n = \operatorname{Li}_w(z)/(1-z)$; constants have generating function $z/(1-z)$.

For $\operatorname{Re} z > 0$, the expansion and primitive have ordinary kernels

$$f_P(z) = P_0(z) + \kappa_P + \sum_{j=1}^{M} \frac{1}{(j-1)!} \int_0^\infty t^{j-1}(e^{-zt}-1) V_j(e^{-t})\, \mathrm{d}t, \tag{5.6}$$

$$\begin{aligned}\mathcal{P}_P(z) = {} & \sum_{j=1}^{M} p_j(0)(F_j(z)-F_j(1)) + \kappa_P(z-1) \\ & + \sum_{j=1}^{M} \frac{1}{(j-1)!} \int_0^\infty t^{j-1} \left\{ \frac{e^{-t}-e^{-zt}}{t} - (z-1) \right\} V_j(e^{-t})\, \mathrm{d}t.\end{aligned} \tag{5.7}$$

Near $t=0$, $V_j(e^{-t}) = O(t^{-1}(1+|\log t|)^d)$, while the braced differences vanish to first order in t. Near infinity the positive summation indices supply exponential decay. These estimates, uniform on compact subsets of $\operatorname{Re} z > 0$, justify Fubini and all fixed z-derivatives. The normal partial-fraction formula gives the continuation to the larger slit plane without imposing an unjustified integral convergence claim there.

This construction links three operations under one normalization: spatial differentiation multiplies the Mellin kernel by powers of $-t$; spectral differentiation of the harmonic Dirichlet series inserts logarithms of the distinguished integer; and spatial integration gives the explicit quotient in (5.7). These operations are distinct and their constants have been fixed separately.

6 One singular Laurent coordinate at every Gamma–Appell degree

The newest canonical-digamma report proves that a Gamma–Appell sequence matches the nondecaying harmonic residues of the digamma powers, and asks which lower spectral coordinates survive [3]. The higher-power and mixed-resolvent reports give terminating harmonic-word

compilers and complete fifth-power formulas, but leave the sixth and seventh cases and a uniform preferred spectral support as explicit further targets [2, 4]. The result here settles a precise part of that question at every degree: one normalization removes *all lower singular Laurent coordinates*. The remaining terms have ordinary absolutely convergent multiple-zeta derivatives. It does not remove every lower-depth arithmetic constant, and it makes no assertion of arithmetic independence.

6.1 The normalized moments and their finite data

Define

$$\mathsf{P}_m(Y) = m![t^m]\Gamma(1-t)e^{tY}, \qquad \mathsf{S}_m = \mathrm{FP}\int_0^1 \mathsf{P}_m(\psi(x))\,\mathrm{d}x. \tag{6.1}$$

The finite part uses the original cutoff $x = \varepsilon$. For $m \geq 2$, write

$$\frac{1}{\Gamma(1+t)} = \sum_{j\geq 0} c_j t^j, \qquad \lambda_d = [s^1]Z_d(1+s, \underbrace{1,\dots,1}_{d-1}), \quad d \geq 1. \tag{6.2}$$

Here Z_d is the strict multiple zeta function, with the largest summation index first. The coefficient of a Laurent germ has no additional factorial or alternating sign. In particular $\lambda_1 = -\gamma_1$.

Let $p_{m,j}(n)$ be the coefficient of $(x+n)^{-j}$ in $\mathsf{P}_m(\psi(x))$ at $x = -n$, and put

$$\begin{aligned} e_r(n) &= H_n(\{1\}^r), \qquad e_0(n) = 1, \\ \delta_m(n) &= p_{m,1}(n) + m!e_{m-1}(n), \end{aligned} \tag{6.3}$$

$$\mathsf{J}_m = \sum_{n\geq 1}\sum_{j=2}^{m} \frac{p_{m,j}(n) - p_{m,j}(n-1)}{(j-1)n^{j-1}}, \tag{6.4}$$

$$\mathsf{D}_m = -\sum_{n\geq 1}(\delta_m(n) - \delta_m(n-1))\log n. \tag{6.5}$$

The infinite expressions in (6.4) and (6.5) will be replaced by finite multiple-zeta formulas. They are also useful ordinary convergent representatives for numerical evaluation.

Theorem 6.1 (All lower singular Laurent coordinates disappear)**.** *For every integer $m \geq 2$,*

$$\boxed{\mathsf{S}_m = m!(c_m - \lambda_{m-1}) + \mathsf{J}_m + \mathsf{D}_m.} \tag{6.6}$$

The following assertions specify a finite arithmetic normal form.

1. *J_m is a finite linear combination of ordinary admissible multiple zeta values, with coefficients in $\mathbb{Q}[\zeta(2),\dots,\zeta(m)]$.*
2. *$\mathsf{D}_2 = \mathsf{D}_3 = 0$. For $m \geq 4$, D_m is a finite linear combination of first partial derivatives $\partial_j Z_d(s_1,\dots,s_d)$ at admissible positive-integer indices, with $d \leq m-3$. Each derivative has an ordinary absolutely convergent defining series in a full parameter neighborhood of its evaluation point.*
3. *Every fixed word in these finite expressions has weight at most $m-1$. Assigning weight r to a coefficient $\zeta(r)$ makes each summand have total weight $m-1$ before the spectral differentiation.*

Thus λ_{m-1} is the only singular Laurent coefficient needed by this construction. No lower λ_d is retained.

The finite-part identity. The scalar version of Theorem 2.3 is linear in its polynomial argument. Its large-circle constant for P_m is

$$m![t^m]\Gamma(1-t)\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{it\theta}\,\mathrm{d}\theta = m![t^m]\Gamma(1-t)\frac{\sin \pi t}{\pi t} = m!c_m.$$

The last equality is Euler's reflection formula. The generating identity

$$\operatorname*{Res}_{x=-n}\frac{\Gamma(1-t)\Gamma(x)}{\Gamma(x-t)} = -t\prod_{k=1}^{n}(1+t/k)$$

explains the comparison term $-m!e_{m-1}(n)$; its use here is consistent with the established differential normalization in [3]. Since

$$e_{m-1}(n) - e_{m-1}(n-1) = \frac{e_{m-2}(n-1)}{n},$$

the meromorphic residue-difference series is

$$\sum_{n\geq 1}\frac{p_{m,1}(n)-p_{m,1}(n-1)}{n^s} = -m!Z_{m-1}(1+s,\{1\}^{m-2}) + \sum_{n\geq 1}\frac{\delta_m(n)-\delta_m(n-1)}{n^s}. \tag{6.7}$$

The next proof shows that the second series is holomorphic for $\Re s > -1$ and that its coefficient of s is exactly D_m. Extracting the coefficient in the scalar bridge gives (6.6), including its endpoint and angular constants. The finite-word description of J_m follows from the same finite-difference and quasi-shuffle construction as Theorem 3.1; its initial index is at least two. □

6.2 Why the correction is a finite convergent derivative family

The proof can be made completely explicit without assigning a new regularization to any mixed term. Set

$$h = H_n, \qquad A_r(n) = \zeta(r) - H_n^{(r)} = \sum_{k>n} k^{-r}, \qquad r \geq 2. \tag{6.8}$$

Lemma 6.2 (The tail ideal and the depth bound)**.** *For every fixed m, $\delta_m(n)$ is a polynomial $\delta_m(h, A_2, A_3, \ldots)$ with coefficients in the ordinary zeta algebra. It vanishes identically when all A_r are set to zero. For $m \geq 4$, its degree in the harmonic variables $h, A_2, A_3, \ldots$, counting each of them with degree one, is at most $m-3$. Moreover $\delta_2 = \delta_3 = 0$.*

Proof. The exact local Laurent expansion is

$$\psi(-n+v) = H_n - \gamma - \frac{1}{v} + \sum_{r\geq 1}\bigl(H_n^{(r+1)} + (-1)^{r+1}\zeta(r+1)\bigr)v^r. \tag{6.9}$$

Only finitely many coefficients enter any fixed residue. Substituting $H_n^{(r)} = \zeta(r) - A_r$ proves polynomiality. With every $A_r = 0$, the right side is $h - \gamma - \pi\cot\pi v$. The universal cotangent residue

$$\operatorname*{Res}_{v=0} e^{-\pi t\cot \pi v} = -\frac{\sin \pi t}{\pi}$$

therefore gives the zero-tail residue generator (the residue identity follows coefficientwise from (2.10))

$$-\Gamma(1-t)e^{(h-\gamma)t}\frac{\sin \pi t}{\pi} = -t\frac{e^{(h-\gamma)t}}{\Gamma(1+t)}.$$

On the other hand the zero-tail specialization of the elementary harmonic generator is

$$\sum_{r\geq 0} e_r t^r = \exp\left(ht + \sum_{r\geq 2} \frac{(-1)^{r-1}\zeta(r)}{r} t^r\right) = \frac{e^{(h-\gamma)t}}{\Gamma(1+t)}.$$

Their sum in (6.3) is zero. This proves membership in the ideal generated by the A_r, using exact Gamma identities.

For the harmonic degree, first keep $h_r = H_n^{(r)}$ as independent variables and write $h = h_1$. The centered Gamma–Appell generator is

$$\Gamma(1-t)e^{tY} = \exp\left((Y+\gamma)t + \sum_{r\geq 2} \frac{\zeta(r)}{r} t^r\right).$$

For $m \geq 3$, the only terms of harmonic degree $m-1$ and $m-2$ in its simple-pole coefficient are respectively

$$-mh^{m-1}, \qquad \frac{m!}{2(m-3)!} h_2 h^{m-3}.$$

Indeed the first chooses one $-1/v$. To have only one fewer harmonic factor, one must choose two $-1/v$ factors and one $h_2 v$ factor. Any other positive power of v, additional negative-power factor, or ordinary-zeta coefficient lowers the harmonic degree further. The same two terms occur in $-m!e_{m-1}$ by expansion of the elementary harmonic generator. They cancel in δ_m. Replacing h_r by $\zeta(r) - A_r$ cannot increase this degree. Direct extraction at $m = 2, 3$ gives zero defects, completing the proof. □

Let $h = H_{n-1}$ and $v = 1/n$, while $A_r = A_r(n)$ is the *strict upper* tail. The finite difference is consequently the exact polynomial

$$\delta_m(n) - \delta_m(n-1) = \delta_m(h+v, A_2, A_3, \ldots) - \delta_m(h, A_2 + v^2, A_3 + v^3, \ldots). \tag{6.10}$$

Every monomial has the form

$$a\, v^q h^b \prod_{r\geq 2} A_r^{b_r}, \qquad q \geq 1. \tag{6.11}$$

If it contains no upper tail, then $q \geq 2$. This last restriction is decisive: a term with $q = 1$ arises only by changing h, so the tail-ideal property leaves an upper tail behind.

For a word $\mathbf{a} = (a_1, \ldots, a_r)$ of entries at least two, define

$$T_{\mathbf{a}}(n) = \sum_{k_1 > \cdots > k_r > n} k_1^{-a_1} \cdots k_r^{-a_r}.$$

The empty upper word and empty lower word both have value one. Apply the strict quasi-shuffle rule separately to the upper-tail product and to h^b. Each monomial becomes a finite sum of $n^{-q} T_{\mathbf{a}}(n) H_{n-1}(\mathbf{b})$. Then

$$\sum_{n\geq 1} \frac{T_{\mathbf{a}}(n) H_{n-1}(\mathbf{b})}{n^{q+s}} = Z(\mathbf{a}, q+s, \mathbf{b}). \tag{6.12}$$

The index ordering on the two sides is identical. If the upper word is empty then $q \geq 2$; otherwise the first entry is at least two. All entries are positive integers at $s = 0$, so these series and their first derivatives converge absolutely on a small complex neighborhood. Equivalently, elementary tail estimates in (6.11) give $O(n^{-2}(1+\log n)^B)$ before inserting n^{-s}, proving normal convergence for $\Re s > -1$.

Differentiation of (6.12) therefore gives the explicit conversion rule

$$-\sum_{n\geq 1}\frac{T_{\mathbf{a}}(n)H_{n-1}(\mathbf{b})\log n}{n^q}=\partial_{r+1}Z(\mathbf{a},q,\mathbf{b}),\qquad r=|\mathbf{a}|. \tag{6.13}$$

Taking a nonzero finite difference replaces at least one harmonic factor by a power of v. The remaining harmonic-factor count is at most $m-4$; adjoining the distinguished n in (6.12) gives depth at most $m-3$. This completes the proof of Theorem 6.1 and provides a terminating finite algorithm for every term it retains.

6.3 Explicit sixth and seventh normalized powers

For an admissible integer word $\mathbf{w}$, write $Z_{\mathbf{w}}=Z(\mathbf{w})$ and let

$$\Delta_{ij}Z_{\mathbf{w}}=(\partial_i-\partial_j)Z(\mathbf{w}). \tag{6.14}$$

These are derivatives of a genuinely free-order analytic function. They are not obtained by differentiating an identity valid only at integer orders.

Theorem 6.3 (The sixth normalized digamma moment)**.** *The sixth-degree identity is*

$$\boxed{\mathsf{S}_6=720(c_6-\lambda_5)+245\zeta(2)\zeta(3)+175\zeta(5)+\mathsf{D}_6,} \tag{6.15}$$

where the entire mixed correction is the following finite sum of convergent derivatives:

$$\begin{aligned}\mathsf{D}_6=\;&120\zeta(2)\Delta_{12}Z_{2,1}-60\Delta_{23}Z_{2,2,1}+120\Delta_{12}Z_{3,1,1}\\&-180\Delta_{12}Z_{4,1}-30\Delta_{12}Z_{2,3}+138\zeta'(5)-60\zeta(2)\zeta'(3)+30\zeta(3)\zeta'(2).\end{aligned} \tag{6.16}$$

Theorem 6.4 (The seventh normalized digamma moment)**.** *The seventh-degree identity is*

$$\boxed{\mathsf{S}_7=5040(c_7-\lambda_6)+\frac{3731}{4}\zeta(3)^2+\frac{918631}{240}\zeta(6)+\mathsf{D}_7,} \tag{6.17}$$

where

$$\begin{aligned}\mathsf{D}_7=\;&840\zeta(2)\Delta_{12}Z_{2,1,1}-840\zeta(2)\Delta_{12}Z_{3,1}+210\zeta(2)\partial_1Z_{2,2}\\&-420\Delta_{23}Z_{2,2,1,1}+630\Delta_{23}Z_{2,3,1}+840\Delta_{12}Z_{3,1,1,1}\\&+420\Delta_{12}Z_{3,1,2}+210(2\Delta_{13}+\Delta_{23})Z_{3,2,1}+210\partial_1Z_{2,2,2}\\&+798\partial_1Z_{3,3}-1260\Delta_{12}Z_{4,1,1}+168\Delta_{12}Z_{2,4}+1596\Delta_{12}Z_{5,1}\\&+\frac{651}{2}\zeta(4)\zeta'(2)+336\zeta(2)\zeta'(4)-378\zeta(3)\zeta'(3)-504\zeta'(6).\end{aligned} \tag{6.18}$$

For completely explicit integrands, put $U=\psi(x)+\gamma$. Then

$$\begin{aligned}\mathsf{P}_6(\psi)=\;&U^6+15\zeta(2)U^4+40\zeta(3)U^3+\frac{405}{2}\zeta(4)U^2\\&+(120\zeta(2)\zeta(3)+144\zeta(5))U+40\zeta(3)^2+\frac{2745}{8}\zeta(6),\end{aligned} \tag{6.19}$$

$$\begin{aligned}\mathsf{P}_7(\psi)=\;&U^7+21\zeta(2)U^5+70\zeta(3)U^4+\frac{945}{2}\zeta(4)U^3\\&+(420\zeta(2)\zeta(3)+504\zeta(5))U^2+\left(280\zeta(3)^2+\frac{19215}{8}\zeta(6)\right)U\\&+945\zeta(3)\zeta(4)+504\zeta(2)\zeta(5)+720\zeta(7).\end{aligned} \tag{6.20}$$

6.4 Recovering raw moments without external tables

Put $Q_m = \mathrm{FP}\int_0^1 \psi(x)^m\,\mathrm{d}x$, with $Q_0 = 1$. The inverse of the Gamma–Appell normalization is the finite formula

$$\boxed{Q_m = m!\sum_{k=0}^{m}\frac{(-1)^{m-k}c_{m-k}}{k!}\,\mathsf{S}_k.} \tag{6.21}$$

Indeed, the defining polynomial identity gives the formal-power-series relation

$$\sum_{m\ge0}\frac{\mathsf{S}_m}{m!}t^m = \Gamma(1-t)\sum_{m\ge0}\frac{Q_m}{m!}t^m.$$

Multiplication by $1/\Gamma(1-t) = \sum_{j\ge0}(-1)^j c_j t^j$ proves (6.21). This is a coefficientwise polynomial argument; it requires no interchange of an infinite series with a finite-part integral. The coefficients themselves are determined by

$$c_0 = 1, \qquad jc_j = \gamma c_{j-1} + \sum_{r=2}^{j}(-1)^{r+1}\zeta(r)c_{j-r}, \qquad j \ge 1. \tag{6.22}$$

For completeness, the lower degrees already treated in the earlier reports [2, 3] take the following form in the present normalization. They are recovered prior inputs, included to make the sixth- and seventh-power evaluations self-contained:

$$\mathsf{S}_0 = 1, \qquad \mathsf{S}_1 = \gamma, \tag{6.23}$$
$$\mathsf{S}_2 = 2(c_2 - \lambda_1),$$
$$\mathsf{S}_3 = 6(c_3 - \lambda_2) + 3\zeta(2),$$
$$\mathsf{S}_4 = 24(c_4 - \lambda_3) + 12\zeta(3) + 4\zeta'(3),$$
$$\mathsf{S}_5 = 120(c_5 - \lambda_4) + \frac{260}{3}\zeta(4) + \mathsf{D}_5, \tag{6.24}$$
$$\mathsf{D}_5 = 20\zeta(2)\zeta'(2) - 10\partial_2 Z_{2,2} + 20\Delta_{12}Z_{3,1} - 30\zeta'(4). \tag{6.25}$$

Here $Z_{a,b} = Z_2(a,b)$ and $\Delta_{12} = \partial_1 - \partial_2$, with all derivatives evaluated at the displayed indices. In particular, all terms in D_5 have ordinary absolutely convergent defining series. The value $\mathsf{S}_1 = \gamma$ uses $\mathrm{FP}\int_0^1 \psi(x)\,\mathrm{d}x = 0$, as follows directly from $\int_\varepsilon^1 \psi(x)\,\mathrm{d}x = -\log\Gamma(\varepsilon)$.

Here is a short independent check of the lower arithmetic terms. The Laurent coefficients at degrees two and three give $\mathsf{J}_2 = 0$, $\mathsf{J}_3 = 3\zeta(2)$, and $\delta_2 = \delta_3 = 0$. At degrees four and five,

$$\mathsf{J}_4 = 12Z_{2,1} = 12\zeta(3), \qquad \mathsf{J}_5 = 60Z_{2,1,1} + 10Z_{2,2} + 10\zeta(2)^2 - \frac{35}{6}\zeta(4).$$

The classical identities $Z_{2,1,1} = \zeta(4)$, $Z_{2,2} = 3\zeta(4)/4$, and $\zeta(2)^2 = 5\zeta(4)/2$ give $\mathsf{J}_5 = 260\zeta(4)/3$. In the tail notation (6.8),

$$\delta_4 = -4A_3, \qquad \delta_5 = 5A_2^2 - 20\zeta(2)A_2 - 20hA_3 + 25A_4.$$

For the consecutive difference use $h = H_{n-1}$, $A_r = A_r(n)$, and $v = 1/n$. These two defects give respectively

$$4v^3, \qquad -10A_2v^2 - 20A_3v + 20hv^3 - 30v^4 + 20\zeta(2)v^2.$$

Multiplication by $-\log n$ and summation proves $\mathsf{D}_4 = 4\zeta'(3)$ and (6.25) directly from the convergent strict multiple sums. Equations (6.21)–(6.25), together with (6.15) and (6.17), therefore give explicit finite reductions of Q_6 and Q_7 using only coordinates defined in this article.

6.5 Proof ledger for the two higher-degree identities

For reproducibility, the residue defects that generate every directional term above are printed in full. In the next two formulas $h = H_n$ and $A_r = A_r(n)$ as in (6.8):

$$\delta_6 = 30A_2^2h - 90A_2A_3 - 120\zeta(2)A_2h - 60A_3h^2 + 120\zeta(2)A_3 + 150A_4h - 138A_5, \tag{6.26}$$

$$\begin{aligned}\delta_7 = {} & 70A_2^3 + 105A_2^2h^2 - 315\zeta(2)A_2^2 - 630A_2A_3h + 588A_2A_4 \\ & - 420\zeta(2)A_2h^2 - 126\zeta(4)A_2 + 259A_3^2 - 140A_3h^3 + 840\zeta(2)A_3h - 280\zeta(3)A_3 \\ & + 525A_4h^2 - 651\zeta(2)A_4 - 966A_5h + 833A_6.\end{aligned} \tag{6.27}$$

They follow by finite coefficient extraction in (6.9), (6.1), and the elementary harmonic generator. Applying (6.10) and (6.13) to these two polynomials gives the unsimplified directional formulas recorded in the attached exact coefficient tables.

Only the following free-order identities are needed to simplify their symmetric derivative parts:

$$\zeta(a)\zeta(b) = Z(a,b) + Z(b,a) + \zeta(a+b), \tag{6.28}$$

$$Z(s,s,s) = \frac{\zeta(s)^3 - 3\zeta(s)\zeta(2s) + 2\zeta(3s)}{6}. \tag{6.29}$$

Both hold on open absolute-convergence domains. In particular

$$\begin{aligned}(\partial_1+\partial_2)Z(2,2) &= \zeta(2)\zeta'(2) - \zeta'(4),\\ (\partial_1+\partial_2)Z(3,3) &= \zeta(3)\zeta'(3) - \zeta'(6),\\ (\partial_1+\partial_2+\partial_3)Z(2,2,2) &= \frac{3}{4}\zeta(4)\zeta'(2) - \zeta(2)\zeta'(4) + \zeta'(6).\end{aligned}$$

The derivative identities for $(2,3)$ and $(2,4)$ are obtained by differentiating (6.28) in either argument. Direct substitution yields (6.16) and (6.18). Integer-index duality is used only for ordinary values in J_m, never for these derivatives.

For completeness, the ordinary word calculations before the last reductions are

$$\begin{aligned}\mathsf{J}_6 = {} & 90\zeta(2)\zeta(3) + 60\zeta(2)Z_{2,1} + 360Z_{2,1,1,1} + 60Z_{2,1,2} + 60Z_{2,2,1} \\ & - 30Z_{2,3} - 15\zeta(2)\zeta(3) - 15Z_{3,2} - 35Z_{4,1} - \frac{5}{2}\zeta(5),\end{aligned} \tag{6.30}$$

$$\begin{aligned}\mathsf{J}_7 = {} & 588\zeta(4)\zeta(2) + 630\zeta(3)Z_{2,1} + 420\zeta(2)Z_{2,1,1} + 2520Z_{2,1,1,1,1} \\ & + 420(Z_{2,1,1,2} + Z_{2,1,2,1} + Z_{2,2,1,1}) - 210(Z_{2,1,3} + Z_{2,3,1}) - 42Z_{2,4} \\ & - 42\zeta(3)^2 - 105\zeta(2)Z_{3,1} - 105(Z_{3,1,2} + Z_{3,2,1}) - 63Z_{3,3} \\ & - 77\zeta(2)\zeta(4) - 245Z_{4,1,1} - 77Z_{4,2} - \frac{35}{2}Z_{5,1} + \frac{189}{10}\zeta(6).\end{aligned} \tag{6.31}$$

The familiar weight-five Euler and duality identities reduce (6.30) to $245\zeta(2)\zeta(3) + 175\zeta(5)$. The weight-six entries needed for (6.31) are as follows. Write $a = \zeta(3)^2$, $b = \zeta(6)$.

Admissible value or equal pair	Reduction
$Z_{2,1,1,1,1}$	b
$Z_{2,1,1,2} = Z_{2,4}$	$25b/12 - a$
$Z_{2,1,2,1} = Z_{3,3}$	$(a-b)/2$
$Z_{2,2,1,1} = Z_{4,2}$	$a - 4b/3$
$Z_{2,1,3}$	$a - 13b/16$
$Z_{2,3,1} = Z_{3,1,2}$	$53b/24 - 3a/2$
$Z_{3,2,1}$	$3a - 203b/48$
$Z_{4,1,1}$	$23b/16 - a$
$Z_{5,1}$	$3b/4 - a/2$

These are ordinary classical multiple-zeta reductions, not new claims of special-value priority. One finite derivation uses the strict stuffle product, the binary iterated-integral shuffle product, and reversal-complement duality [12]. At weight six, take the products of admissible words of weights $2+4$ and $3+3$, their reversals, dualities, and the regularized shuffle-minus-stuffle relation of (1) with each admissible weight-five word. Along with $Z(6) = \zeta(6)$, this gives 49 explicitly generated equations in 16 admissible weight-six words. The provided rational certificate contains 16 independent rows and their inverse; multiplication checks the resulting table exactly. Substitution in (6.31), using $\zeta(2)\zeta(4) = 7\zeta(6)/4$, gives the ordinary constant in (6.17). This also specifies a short independent way of auditing its less transparent rational numerator.

6.6 Validation and the remaining arithmetic question

The accompanying program constructs the pole coefficients, the tail polynomials, their finite differences, and the distinguished-index word maps using exact rational algebra. It checks the weight-six reduction certificate and the directional simplifications separately. The analytic statements above justify all infinite operations; a finite symbolic replay is not a substitute for those statements.

An independent numerical diagnostic compares the locally subtracted integral of the explicit polynomial (6.19) or (6.20) with a different representation. In the latter, $m!(c_m - \lambda_{m-1})$ is evaluated through the ordinary Gamma-quotient beta kernel established in [3]:

$$m!(c_m - \lambda_{m-1}) = -\int_0^1 (-\log y)^m \frac{-\log(1-y) - y}{(1-y)^2[-\log(1-y)]^2}\, \mathrm{d}y. \tag{6.32}$$

The correction (6.5) is evaluated from its tail polynomial at $n \le 96$, with the remaining ordinary sum accelerated by 42 inverse-power terms and Hurwitz-zeta order derivatives. At 80 working decimal digits the comparisons were

Moment	Direct numerical value	Absolute discrepancy
S_6	$20.822613154257822426734055105133 3\ldots$	8.80×10^{-56}
S_7	$123.619906884273853180167345465232\ldots$	5.73×10^{-54}

These are floating-point diagnostics, not interval certificates. They test the finite-part endpoint terms and the ordinary constant reductions through analytically different formulas.

Theorems 6.1–6.4 remove an ambiguity from the source research question. A single singular Laurent coordinate is sufficient in every degree, while additional *convergent* derivative coordinates already occur at low degree. The next arithmetic problem is to determine relations among the explicitly retained directions in (6.16) and (6.18), within a stated relation family or a stated algebra of periods. The depth bound is a constructive upper bound; it is not a proof that these directions are necessary or independent.

7 Every order derivative on the odd Lerch lattice

The report *Continuous Depth Lerch* asks in its final question Q8 for the spectral derivatives of its continuous-depth family at odd positive integers [5]. Its formula `cdl:eq:oddF` gives the values there, but does not permit differentiation in the discrete index. We resolve that question in the precise form requested: an explicit finite formula for every order derivative and every mixed depth derivative, including the reflected spectral coordinates. We also give normally convergent series for all those coordinates. Thus the answer includes a constructive evaluation procedure at the negative reflected orders, where the original alternating series need not converge.

The analytic inputs are the source report's proved centered reflection formula `cdl:eq:completed`, Euler's classical transformation of series, and the usual Gamma and Hurwitz-zeta identities [11, 8, 9]. The new claim is the following continuation of the inspected source question, not priority for those classical inputs or for the special cases recovered below. Euler's transformation is recorded in DLMF §3.9(ii); the normally convergent, parameter-differentiated use needed here is proved explicitly.

7.1 Functions, branches, and normalization

For $\operatorname{Re}\rho > 0$, put

$$f_\rho(t) = (2\cosh(t/2))^{-\rho}, \qquad t \geq 0, \tag{7.1}$$

$$Z_\rho(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} f_\rho(t)\, \mathrm{d}t, \qquad \operatorname{Re} s > 0. \tag{7.2}$$

The logarithm in f_ρ is real on the positive real axis. The function $Z_\rho(s)$ is entire in s: subtracting finitely many Taylor terms of f_ρ near zero gives its meromorphic Mellin transform, and the zeros of $1/\Gamma(s)$ cancel its poles. The same subtractions on compact parameter sets establish joint holomorphy for $\operatorname{Re}\rho > 0$. In the absolutely convergent region $\operatorname{Re} s > \operatorname{Re}\rho$,

$$Z_\rho(s) = \sum_{k=0}^{\infty} \frac{(-1)^k (\rho)_k}{k!(k+\rho/2)^s}. \tag{7.3}$$

Powers of $k + \rho/2$ use the logarithm on the right half-plane.

Define

$$b_\rho(x) = \frac{(2\pi)^{\rho-2}}{\Gamma(\rho)} \Gamma\left(\frac{\rho}{2} + \frac{ix}{2\pi}\right) \Gamma\left(\frac{\rho}{2} - \frac{ix}{2\pi}\right), \tag{7.4}$$

$$F_\rho(s) = \frac{1}{\Gamma(s)} \int_0^\infty x^{s-1} b_\rho(x)\, \mathrm{d}x, \qquad \operatorname{Re} s > 0. \tag{7.5}$$

This is $\mathcal{F}_{\rho,1/2}(s;0)$ in the continuous-depth report. Its entire continuation follows from the same Mellin-subtraction argument. For complex ρ, the Gamma product is a product of two functions, not a squared modulus.

The centered reflection identity is

$$F_\rho(s) = \frac{\pi(2\pi)^{\rho+s-2}}{\Gamma(s)\sin(\pi s/2)} Z_\rho(1-s). \tag{7.6}$$

Apparent singularities are removed in the combined expression. For completeness, the beta integral gives the classical Fourier formula [13] $\widehat{f_\rho}(v) = \Gamma(\rho/2 + iv)\Gamma(\rho/2 - iv)/\Gamma(\rho)$, where the Fourier transform is $\int_{\mathbb{R}} e^{ivt} f_\rho(t)\, \mathrm{d}t$. On $0 < \operatorname{Re} s < 1$, insert a decaying exponential in the cosine Mellin integral, use Fubini, and then dominated convergence to obtain

$$\int_0^\infty v^{s-1} \widehat{f_\rho}(v)\, \mathrm{d}v = 2\Gamma(s)\cos(\pi s/2) \int_0^\infty t^{-s} f_\rho(t)\, \mathrm{d}t.$$

The regulator can be removed on the left because the Gamma product decays exponentially. Rescaling gives $F_\rho(s) = 2(2\pi)^{\rho+s-2}\Gamma(1-s)\cos(\pi s/2) Z_\rho(1-s)$; Euler reflection for Gamma gives (7.6). Entire continuation gives the stated global interpretation. This reproduces the source input rather than replacing it with a formal exchange at a pole.

We use ∂_s^q for order derivatives and ∂_ρ^r for depth derivatives. The complete exponential Bell polynomials are fixed by

$$\exp\left(\sum_{j\geq 1} d_j \frac{v^j}{j!}\right) = \sum_{q\geq 0} \mathcal{B}_q(d_1, \ldots, d_q) \frac{v^q}{q!}, \qquad \mathcal{B}_0 = 1. \tag{7.7}$$

Write $H_N^{(j)} = \sum_{k=1}^N k^{-j}$ and $H_0^{(j)} = 0$.

7.2 The finite formula for all mixed jets

Theorem 7.1 (Odd-lattice order and depth derivatives)**.** *Let $m, r, q \geq 0$ be integers, put $N = 2m$, $n = N + 1$, and set*

$$L = \log(2\pi), \qquad C_{m,\rho} = \frac{(-1)^m \pi (2\pi)^{\rho+N-1}}{N!}, \tag{7.8}$$

$$\kappa_1 = L + \gamma - H_N,$$

$$\kappa_j = (j-1)! \left[\left(1 - \frac{1 + (-1)^j}{2^j} \right) \zeta(j) + (-1)^j H_N^{(j)} \right], \qquad j \geq 2. \tag{7.9}$$

For every $\operatorname{Re} \rho > 0$*,*

$$\boxed{\partial_\rho^r \partial_s^q F_\rho(n) = C_{m,\rho} \sum_{j=0}^{q} (-1)^j \binom{q}{j} \mathcal{B}_{q-j}(\kappa_1, \ldots, \kappa_{q-j}) (\partial_\rho + L)^r \partial_s^j Z_\rho(-N).} \tag{7.10}$$

The operator on the right is explicitly $\sum_{h=0}^{r} \binom{r}{h} L^{r-h} \partial_\rho^h$*. No differentiation with respect to m occurs.*

Proof. The pole-free local identity from which every derivative follows is

$$F_{\rho+u}(n+v) = C_{m,\rho} e^{Lu} R_m(v) Z_{\rho+u}(-N-v), \qquad R_m(v) = e^{Lv} \frac{\Gamma(n)}{\Gamma(n+v)} \sec(\pi v/2). \tag{7.11}$$

It holds for u sufficiently small and $|v| < 1$, by substituting in (7.6) and using $\sin(\pi(n+v)/2) = (-1)^m \cos(\pi v/2)$. All factors in (7.11) are holomorphic there.

The Gamma contribution to the jth logarithmic derivative of R_m is $-\psi^{(j-1)}(n)$. For $j \geq 2$,

$$\psi^{(j-1)}(n) = (-1)^j (j-1)! (\zeta(j) - H_N^{(j)}).$$

The product for cosine gives the convergent expansion

$$\log \sec(\pi v/2) = \sum_{k \geq 1} \frac{(1 - 2^{-2k}) \zeta(2k)}{k} v^{2k}, \qquad |v| < 1.$$

Combining these expansions gives exactly (7.9) and $R_m^{(a)}(0) = \mathcal{B}_a(\kappa_1, \ldots, \kappa_a)$. Now apply the ordinary multivariate Leibniz rule to the holomorphic identity (7.11). The argument $-N - v$ supplies $(-1)^j$, while e^{Lu} supplies $(\partial_\rho + L)^r$. This proves (7.10) with every combinatorial coefficient fixed. □

For example, putting $Z^{[j]} = \partial_s^j Z_\rho(-N)$, the first two order derivatives are

$$F'_\rho(n) = C_{m,\rho} (\kappa_1 Z^{[0]} - Z^{[1]}), \tag{7.12}$$

$$F''_\rho(n) = C_{m,\rho} ((\kappa_1^2 + \kappa_2) Z^{[0]} - 2\kappa_1 Z^{[1]} + Z^{[2]}), \tag{7.13}$$

$$\kappa_2 = \tfrac{1}{2}\zeta(2) + H_N^{(2)}, \qquad \kappa_3 = 2(\zeta(3) - H_N^{(3)}).$$

To recover the already known value, define $U_m(\rho) = [z^{2m}] \sec(z)^\rho$. The even Taylor expansion of f_ρ gives

$$Z_\rho(-2m) = (-1)^m (2m)! 2^{-\rho-2m} U_m(\rho), \qquad F_\rho(2m+1) = \frac{\pi^{\rho+2m}}{2} U_m(\rho). \tag{7.14}$$

These are the $q = 0$ inputs. The terms $Z^{[j]}$ with $j > 0$ in (7.10) are additional data; discarding them would give an incorrect spectral derivative.

7.3 A convergent elementary series for every reflected jet

The next theorem makes the reflected coordinates in (7.10) independently computable. In particular, it avoids using the reflection equation itself as a numerical definition of the coordinates that are supposed to verify that equation.

Theorem 7.2 (Normally convergent Euler series at every order)**.** *For* $\operatorname{Re}\rho > 0$ *and* $s \in \mathbb{C}$*, define*

$$A_k(\rho, s) = \frac{(\rho)_k}{k!(k+\rho/2)^s}, \qquad D_\ell(\rho, s) = \sum_{k=0}^{\ell} (-1)^k \binom{\ell}{k} A_k(\rho, s). \tag{7.15}$$

Then

$$\boxed{Z_\rho(s) = \sum_{\ell=0}^{\infty} \frac{D_\ell(\rho, s)}{2^{\ell+1}}.} \tag{7.16}$$

This series is absolutely and locally normally convergent throughout $\{\operatorname{Re}\rho > 0\} \times \mathbb{C}$*. Every fixed mixed derivative in* ρ, s *is obtained by differentiating its finite inner sums termwise.*

Proof. For $z \in \Omega = \mathbb{C} \setminus [1, \infty)$, initially $\operatorname{Re} s > 0$, let

$$A(\rho, s; z) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\rho t/2} (1 - ze^{-t})^{-\rho} \, \mathrm{d}t. \tag{7.17}$$

Use the principal logarithm of $1 - ze^{-t}$. It is available on Ω, since $1 - ze^{-t}$ cannot lie on the nonpositive real axis there. On compact subsets, the integrand is bounded at zero after the factor t^{s-1} is removed and decays exponentially at infinity. Thus (7.17) is jointly holomorphic in this initial domain. Its power series at $z = 0$ is $\sum_{k \geq 0} A_k(\rho, s) z^k$.

The coefficient identity

$$\left(z \frac{\partial}{\partial z} + \frac{\rho}{2} \right) A(\rho, s+1; z) = A(\rho, s; z) \tag{7.18}$$

holds first in $|z| < 1$ and then throughout Ω. Iterating it from $\operatorname{Re}(s + M) > 0$ continues A to every $s \in \mathbb{C}$. The definitions agree on overlaps by the same identity, so this is a joint holomorphic continuation. At $z = -1$, it equals $Z_\rho(s)$.

Set

$$B(\rho, s; v) = \frac{1}{1-v} A\left(\rho, s; \frac{-v}{1-v} \right).$$

If $|v| < 1$, then $-v/(1-v)$ is outside $[1, \infty)$: a point of that cut would require $v \in (1, \infty)$. Hence B is holomorphic for $|v| < 1$. Its Taylor coefficient is $D_\ell(\rho, s)$, by expanding $(1-v)^{-k-1}$ in the defining power series near zero. Taking $v = 1/2$ yields $B(\rho, s; 1/2) = 2Z_\rho(s)$ and proves (7.16).

For a compact parameter set, choose any fixed radius r with $1/2 < r < 1$. Cauchy's coefficient estimate on $|v| = r$ gives $|D_\ell| \leq Cr^{-\ell}$ uniformly on that set. It proves absolute normal convergence of (7.16). Apply the same argument on a slightly enlarged compact parameter neighborhood to justify every fixed mixed derivative. This argument proves convergence even when (7.3) is a divergent ordinary series. □

All finite derivative coefficients admit a harmonic expression. Put $a_k = k + \rho/2$ and

$$h_k^{(j)}(\rho) = \sum_{\ell=0}^{k-1} (\rho + \ell)^{-j}, \qquad d_j = (-1)^{j-1}(j-1)! \left(h_k^{(j)}(\rho) - \frac{s}{2^j a_k^j} \right).$$

Then $\partial_\rho^r A_k = A_k \mathcal{B}_r(d_1, \dots, d_r)$. For mixed derivatives a particularly safe finite generating formula is

$$\frac{A_k(\rho+u, s+v)}{A_k(\rho, s)} = e^{-v \log a_k} \prod_{\ell=0}^{k-1} \left(1 + \frac{u}{\rho+\ell}\right) \left(1 + \frac{u}{2a_k}\right)^{-s-v}. \tag{7.19}$$

Only the requested bidegree is needed. This also distinguishes a mixed derivative from the product of two independent logarithmic derivatives.

At the reflected integer $s = -N$, $N \geq 0$, the formula is especially explicit:

$$\begin{aligned} \partial_\rho^r \partial_s^q A_k(\rho, -N) = (-1)^q r! \frac{(\rho)_k}{k!} a_k^N [u^r] \prod_{\ell=0}^{k-1} \left(1 + \frac{u}{\rho+\ell}\right) \left(1 + \frac{u}{2a_k}\right)^N \\ \cdot \left(\log a_k + \log\left(1 + \frac{u}{2a_k}\right)\right)^q. \end{aligned} \tag{7.20}$$

For positive rational ρ, every inner coefficient here is a rational linear combination of powers of logarithms of positive rational numbers. Equations (7.20), (7.16), and (7.10) therefore constitute a complete ordinary convergent series evaluation of all the reflected mixed jets in the theorem. The outer Euler series need not terminate. At a positive integer ρ and $q = r = 0$, its inner sequence is polynomial at negative integer s, so finite differences do terminate; differentiation can remove that termination.

7.4 Independent Gamma-moment realization

The positive-half-line Gamma integral gives a different exact representation for checking the lattice formulas. Put

$$a_j(\rho, x) = \delta_{j1} \log(2\pi) - \psi^{(j-1)}(\rho) + 2^{-j} \left\{ \psi^{(j-1)}\left(\frac{\rho}{2} + \frac{ix}{2\pi}\right) + \psi^{(j-1)}\left(\frac{\rho}{2} - \frac{ix}{2\pi}\right) \right\}. \tag{7.21}$$

For the order derivatives define $c_1(x; s) = \log x - \psi(s)$ and $c_j(x; s) = -\psi^{(j-1)}(s)$ for $j \geq 2$. Direct differentiation of (7.5) gives, for $\operatorname{Re} s > 0$,

$$\begin{aligned} \partial_\rho^r \partial_s^q F_\rho(s) = \frac{1}{\Gamma(s)} \int_0^\infty x^{s-1} b_\rho(x) \mathcal{B}_r(a_1(\rho, x), \dots, a_r(\rho, x)) \\ \cdot \mathcal{B}_q(c_1(x; s), \dots, c_q(x; s)) \, \mathrm{d}x. \end{aligned} \tag{7.22}$$

This is an ordinary absolutely convergent integral. Near zero, only fixed powers of $\log x$ occur. At infinity, the Gamma product decays as $e^{-x/2}$ times a power of x; its parameter derivatives introduce only logarithmic or polynomial factors on compact parameter sets. These observations justify every differentiation. The left side of (7.10) can therefore be evaluated through (7.22) while its right side is evaluated through the independent Euler series.

7.5 Stieltjes coordinates at depth two

Let the generalized Stieltjes constants have the convention

$$\zeta(1+v, a) = \frac{1}{v} + \sum_{j \geq 0} \frac{(-1)^j \gamma_j(a)}{j!} v^j, \qquad \operatorname{Re} a > 0.$$

An upper parenthesized index on γ_j below is an argument derivative. Set $E_a(s) = s\zeta(s+1, a)$, with the removable value $E_a(0) = 1$.

Proposition 7.3 (Stieltjes transport to every positive order)**.** *For integers $n \geq 1, q \geq 0$, put $N = n - 1$ and $\lambda_j = (-1)^j (j-1)! H_N^{(j)}$. Then*

$$\boxed{E_a^{[q]}(n) = \frac{(-1)^n}{N!} \sum_{j=0}^{q} (-1)^j \binom{q}{j} \mathcal{B}_{q-j}(\lambda_1, \ldots, \lambda_{q-j}) \gamma_j^{(n)}(a).} \tag{7.23}$$

In particular,

$$E_a^{[q]}(1) = (-1)^{q+1} \gamma_q'(a). \tag{7.24}$$

At $a = 1/2$, this gives $\partial_s^q F_2(n)$.

Proof. The shift derivative of Hurwitz zeta gives $\partial_a^n \zeta(1+v, a) = (-1)^n (1+v)_n \zeta(n+1+v, a)$. The pole $1/v$ is independent of a, so

$$E_a(n+v) = \frac{(-1)^n}{(1+v)_N} \sum_{j \geq 0} \frac{(-1)^j \gamma_j^{(n)}(a)}{j!} v^j. \tag{7.25}$$

The logarithmic Taylor coefficients of $N!/(1+v)_N$ are λ_j. Coefficient extraction proves (7.23). Finally, $b_2(x) = x/(2\sinh(x/2))$, so its Mellin transform gives $F_2(s) = s\zeta(s+1, 1/2)$ directly. This proves the last assertion. □

This proposition is an explicit form of the familiar parameter-derivative tower. Its role here is to transport the new lattice-jet formula into the repository's Stieltjes convention; it is not presented as a new Hurwitz-zeta identity. For instance, with $N = 2m$,

$$F_2'(2m+1) = \frac{\gamma_1^{(2m+1)}(1/2) + H_{2m} \gamma_0^{(2m+1)}(1/2)}{(2m)!}.$$

Together with (7.12), this is an exact relation between negative reflected Z_2 order derivatives and positive-order Stieltjes argument derivatives.

7.6 Gamma-weighted polylogarithm order derivatives

Put

$$\begin{aligned} G_\rho(x) &= \Gamma\left(\frac{\rho}{2} + \frac{ix}{2\pi}\right) \Gamma\left(\frac{\rho}{2} - \frac{ix}{2\pi}\right), \\ P_s(x) &= \frac{\mathrm{Li}_s(ie^{-x/2}) - \mathrm{Li}_s(-ie^{-x/2})}{i}, \\ I_\rho(s) &= \int_{\mathbb{R}} G_\rho(x) P_s(x) \, \mathrm{d}x. \end{aligned} \tag{7.26}$$

The polylogarithms use their principal branches. Their arguments stay off the positive real cut. The following formula covers all independent order derivatives, including orders for which the integrand is no longer an elementary logarithm or a classical fixed-order polylogarithm.

Theorem 7.4 (All polylogarithmic order jets under a Gamma weight)**.** *For $\operatorname{Re}\rho > 0$, the integral $I_\rho(s)$ is entire in s, jointly holomorphic in its parameters, and*

$$I_\rho(s) = \Gamma(\rho)(2\pi)^{2-\rho} 2^{1-s} F_{\rho+1}(s) = \frac{4\Gamma(\rho)\pi^{s+2}}{\Gamma(s)\sin(\pi s/2)} Z_{\rho+1}(1-s), \tag{7.27}$$

with all apparent poles removed in the combined expression. For $N = 2m$, $n = N + 1$, replace only the first cumulant in (7.9) by $\widetilde{\kappa}_1 = \log \pi + \gamma - H_N$, and leave $\widetilde{\kappa}_j = \kappa_j$ for $j \geq 2$. Then

$$\partial_s^q I_\rho(n) = \frac{4(-1)^m \Gamma(\rho)\pi^{n+2}}{N!} \sum_{j=0}^{q} (-1)^j \binom{q}{j} \mathcal{B}_{q-j}(\widetilde{\kappa}_1, \ldots, \widetilde{\kappa}_{q-j}) \partial_s^j Z_{\rho+1}(-N). \tag{7.28}$$

Every mixed depth derivative follows by the finite rule

$$\partial_\rho^r \{\Gamma(\rho) Y(\rho+1)\} = \Gamma(\rho) \sum_{h=0}^{r} \binom{r}{h} \mathcal{B}_{r-h}(\psi(\rho), \psi'(\rho), \ldots) Y^{(h)}(\rho+1). \tag{7.29}$$

All these derivatives pass through the ordinary integral in (7.26).

Proof. The root filter in the continuous-depth report, valid first by a convergent series and then by analytic continuation, is

$$\mathcal{F}_{1,1/2}(s; -x) = 2^{s-1} e^{-x/2} P_s(x).$$

Convolve this with $K_\rho(x) = (2\pi)^{\rho-2} e^{x/2} G_\rho(x)/\Gamma(\rho)$. The source's entire-order addition law gives $F_{\rho+1}(s)$ and therefore the first equality in (7.27). The second is (7.6), with the powers of 2 and π collected. The ordinary integral is absolutely convergent locally uniformly in all the stated parameters. One explicit justification uses the vertical contour for $\mathcal{F}_{1,1/2}$ on any line $1/2 < c < 1$, giving $|P_s(x)| \leq Ce^{(1/2-c)x}$ on compact order sets. The Gamma weight then decays integrably at both ends. Fixed order derivatives add powers of the logarithm of the contour variable, and depth derivatives of the Gamma weight add only polynomial or logarithmic factors. The same integrable majorants persist on slightly larger compact sets, proving joint holomorphy and passage of all derivatives.

The local germ at $s = n+v$ now contains $e^{v \log \pi}$ in place of $e^{v \log(2\pi)}$. The proof of Theorem 7.1 gives (7.28). Finally, the logarithmic derivatives of $\Gamma(\rho)$ are $\psi^{(j-1)}(\rho)$, so Bell's formula and Leibniz give (7.29). □

On the integral side of (7.29), the weight replacing G_ρ is

$$G_\rho(x) \mathcal{B}_r(g_1(\rho, x), \ldots, g_r(\rho, x)), \qquad g_j(\rho, x) = 2^{-j} \left\{ \psi^{(j-1)}\left(\frac{\rho}{2} + \frac{ix}{2\pi}\right) + \psi^{(j-1)}\left(\frac{\rho}{2} - \frac{ix}{2\pi}\right) \right\}.$$

Thus the theorem is a family of explicit ordinary integrals containing polygamma factors and spectral derivatives of polylogarithms. Theorem 7.2 evaluates their reflected coordinates by convergent series, including the mixed derivatives appearing in (7.29).

At $\rho = 1$ and $s = 1$, $G_1(x) = \pi \operatorname{sech}(x/2)$. The first derivative has the concrete forms

$$\int_{\mathbb{R}} \frac{\pi}{\cosh(x/2)} \, \partial_s P_s(x)|_{s=1} \, \mathrm{d}x = 2\pi\gamma_1'(1/2) - \pi^3 \log 2 \tag{7.30}$$

$$= \pi^3 \left(\gamma + \log \pi + \frac{4}{3} \log 2 + 12\zeta'(-1) \right). \tag{7.31}$$

Indeed, the first equality follows from $I_1(s) = 2\pi 2^{1-s} E_{1/2}(s)$ and (7.24). For the second use $Z_2(s) = \eta(s-1)$, $\eta(s) = (1 - 2^{1-s})\zeta(s)$, $Z_2(0) = 1/4$, and $Z_2'(0) = -\log 2/3 - 3\zeta'(-1)$ in (7.28). The second order derivative is

$$\int_{\mathbb{R}} \frac{\pi}{\cosh(x/2)} \, \partial_s^2 P_s(x)\Big|_{s=1} \, \mathrm{d}x = -2\pi\gamma_2'(1/2) - 4\pi \log 2 \, \gamma_1'(1/2) + \pi^3 \log^2 2. \tag{7.32}$$

These formulas state the normalization explicitly; in particular, the 2^{1-s} factor cannot be held constant during order differentiation.

7.7 What this resolves and what remains

Theorem 7.1 resolves Q8 in the source's requested Gamma–Stieltjes–Z_ρ-jet form. Theorem 7.2 supplies a convergent evaluator and explicit harmonic coefficient calculus for every reflected coordinate, while Theorem 7.4 transports the result to ordinary Gamma-weighted polylogarithm integrals. This does not assert that arbitrary noninteger-depth spectral jets reduce to a finite algebra of classical zeta and Gamma values. Such a further reduction would be a different arithmetic statement.

No false theorem was found in the inspected continuous-depth report. Its caution that one cannot differentiate a formula known only at integer m is correct. The necessary addition is a holomorphic local identity, (7.11), together with the extra reflected jets. The tilted report already treats independent shift and order derivatives of its matched convolution family; those results are retained as prior work and are not recounted as discoveries here.

Two concrete questions remain. First, determine whether selected rational noninteger depths admit additional finite reductions of $\partial_s^q Z_\rho(-2m)$, specifying the comparison algebra in advance. Second, develop a cancellation-aware implementation of the Euler sums with proved arithmetic error bounds. The analytic theorem supplies normal convergence, while finite floating-point agreement by itself does not provide a certificate for a transcendental identity.

8 Optimal initial moments for adjacent even tail orders

The incoming canonical-digamma report [3] conjectures that the first q centered moments of two unequal even harmonic tails have full formal rank q (its Conjecture `conj:firstq`). That report proves the claim when the two orders coincide and gives a larger sufficient prefix for every unequal pair. We prove the conjectured optimal prefix for the entire next family

$$(p, q) = (2b - 2, 2b), \qquad b \geq 2.$$

The proof also gives an exact determinant with a strictly positive rational recurrence. Thus the nonvanishing is an all-order theorem, independent of numerical experiments. The conjecture for general larger gaps remains open here.

8.1 The inherited coordinates and the exact target

For even integers $2 \leq p \leq q = 2b$, put

$$A_p(n) = \zeta(p) - H_n^{(p)}, \qquad D_{p,q}(s) = \sum_{n \geq 0} A_p(n) A_q(n) (n + \tfrac{1}{2})^{-s}, \qquad D_r = D_{p,q}(-2r).$$

The series initially defines $D_{p,q}$ for $\Re s > 3 - p - q$; the notation D_r uses its meromorphic continuation. The centered asymptotic series for the product of the two tails contains only even powers, so every point $s = -2r$ is regular. This regularity and the following finite coordinate formulas are inherited from the two-tail calculus, not additional conclusions of the rank argument.

The source supplies blocks C_j with mutually inverse transforms

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(\tfrac{1}{2}) C_j, \tag{8.1}$$

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r. \tag{8.2}$$

Here $B_n(x)$ is the Bernoulli polynomial and $B_n = B_n(0)$, with $B_1 = -1/2$. The first b blocks $C_0, \dots, C_{b-1}$ are kept as specified coordinate labels. All subsequent blocks have the form

$$C_j = \sum_{k=0}^{b-1} a_{jk} Z_k, \qquad Z_k = \zeta(2k), \quad Z_0 = -\tfrac{1}{2}, \qquad j \geq b. \tag{8.3}$$

For $d = 2j + 1$ and $e = d - p - q + 2k + 1$, the exact rational coefficients are

$$a_{jk} = \begin{cases} B_e \left\{ \dfrac{\binom{d-q+1}{p-2k}}{d-q+1} + \dfrac{\binom{d-p+1}{q-2k}}{d-p+1} \right\}, & e \geq 0, \\ 0, & e < 0. \end{cases} \tag{8.4}$$

A binomial summand is omitted unless its lower index is at least two and no larger than its upper index. In particular a putative lower index zero is omitted. This endpoint convention is essential to the proof and to the determinant.

We consider the $b \times b$ late-block matrix

$$M_{p,q} = (a_{b+r-1,k})_{1 \leq r \leq b,\ 0 \leq k < b}. \tag{8.5}$$

The $2b \times 2b$ coefficient matrix of $D_0, \dots, D_{2b-1}$ on the formal labels

$$C_0, \dots, C_{b-1}, Z_0, \dots, Z_{b-1}$$

is nonsingular exactly when $M_{p,q}$ is nonsingular. Indeed the corresponding block matrix for $C_0, \dots, C_{2b-1}$ is $\operatorname{diag}(I_b, M_{p,q})$, and the transform (8.1) is triangular with nonzero diagonal $1/(2r+1)$. These are formal ranks over $\mathbb{Q}$; they imply generation statements for the actual values without asserting their arithmetic independence.

8.2 A positive recurrence and the determinant

Define positive rational numbers

$$\alpha_n = \frac{2|B_{2n}|}{(2n)!} = \frac{4\zeta(2n)}{(2\pi)^{2n}}, \qquad n \geq 1. \tag{8.6}$$

The equality follows from Euler's evaluation of even zeta values and establishes positivity at every index. Define $\rho_1 = 1$ and, for $r \geq 1$,

$$\rho_{r+1} = \frac{1}{4(2r+1)^2} \sum_{s=1}^{r} \alpha_{r+1-s} \left\{ 2r(2r+1) + \mathbf{1}_{s \geq 2}\, 2s(2s-1) \right\} \rho_s. \tag{8.7}$$

Every coefficient in this recurrence is strictly positive. In particular $\rho_s > 0$ for every s. The beginning is

$$\rho_1 = 1, \qquad \rho_2 = \frac{1}{36}, \qquad \rho_3 = \frac{11}{5400}, \qquad \rho_4 = \frac{113}{705600}, \qquad \rho_5 = \frac{2509}{190512000}.$$

Theorem 8.1 (The optimal prefix for adjacent even orders)**.** *For $q = 2b \geq 4$ and $p = q - 2$, the matrix in (8.5) has determinant*

$$\boxed{\det M_{q-2,q} = \frac{(-1)^{b(b-1)/2}}{b!} \big((2b+1)!!\big)^2 \rho_{b+1} \neq 0.} \tag{8.8}$$

Consequently the first q moments $D_0, \dots, D_{q-1}$ have full formal rank q, and

$$\operatorname{span}_{\mathbb{Q}}\{D_0, D_1, \dots\} = \operatorname{span}_{\mathbb{Q}}\{D_0, \dots, D_{q-1}\} = \operatorname{span}_{\mathbb{Q}}\{C_0, \dots, C_{b-1}, Z_0, \dots, Z_{b-1}\}. \tag{8.9}$$

The complete first-q moment matrix, in the displayed coordinate order, has determinant

$$\frac{(-1)^{b(b-1)/2}\big((2b+1)!!\big)^2\rho_{b+1}}{b!\,(4b-1)!!}. \tag{8.10}$$

No prefix of fewer than q rows can have rank q on these q formal labels, so the prefix length is optimal in this specified formal system.

8.3 Proof by rotation of the local differential equation

We reproduce the source's differential description so that the argument can be checked directly against (8.4). Put $h(t)=\coth(t/2)$. For $0\le k<b$, define the analytic germs

$$\begin{aligned} f_k(t) &= \frac{1}{2(p-2k)!(q-1)!}\int_0^t u^{p-2k}(t-u)^{q-1}h(u)\,du \\ &\quad+\frac{1}{2(q-2k)!(p-1)!}\int_0^t u^{q-2k}(t-u)^{p-1}h(u)\,du. \end{aligned} \tag{8.11}$$

The first summand is omitted when $p-2k<2$. All included integrands are regular at $u=0$. Their power series converge in $|t|<2\pi$. Expanding

$$h(u)=2\sum_{n\ge0}\frac{B_{2n}}{(2n)!}u^{2n-1}$$

and integrating monomials shows that the coefficient of $t^{2j+1}/(2j+1)!$ in f_k is precisely a_{jk} for $j\ge b$.

For $v=(v_0,\dots,v_{b-1})$ set

$$f_v=\sum_{k=0}^{b-1}v_k f_k,\qquad Q(t)=\sum_{k=0}^{b-1}v_k\frac{t^{q-2k}}{(q-2k)!}.$$

When $p=q-2$, differentiating (8.11) q times gives

$$2f_v^{(q)}(t)=\big(Q''(t)-Q''(0)\big)h(t)+\big(Q(t)h(t)\big)''=:\mathcal{L}_Q(t). \tag{8.12}$$

The subtracted constant removes exactly the omitted lower-index-zero term. Each side is an odd analytic germ at zero.

Rotate $t=ix$, put

$$R(x)=Q(ix)=\sum_{s=1}^{b}r_s x^{2s},\qquad r_s=\frac{(-1)^s}{(2s)!}v_{b-s},\qquad c(x)=\cot(x/2).$$

Because $h(ix)=-ic(x)$ and two differentiations contribute a factor -1, (8.12) gives

$$\mathcal{L}_Q(ix)=i\big((R''-R''(0))c+(Rc)''\big). \tag{8.13}$$

The benefit of this rotation is the one-sided coefficient expansion

$$c(x)=\frac{2}{x}-\sum_{n\ge1}\alpha_n x^{2n-1},\qquad |x|<2\pi, \tag{8.14}$$

where every α_n is positive.

Let Y_r be the ordinary coefficient of x^{2r-1} in the bracket on the right of (8.13). Write $r_{b+1} = 0$ because R has degree at most $2b$. Direct differentiation of the monomials in (8.14) gives, for $1 \le r \le b$,

$$Y_r = d_{r+1} r_{r+1} - \sum_{s=1}^{r} w_{rs} r_s, \tag{8.15}$$

where

$$d_{r+1} = 4(2r+1)^2, \qquad w_{rs} = \alpha_{r+1-s}\,\{2r(2r+1) + \mathbf{1}_{s\ge 2}\, 2s(2s-1)\} > 0. \tag{8.16}$$

For clarity, the coefficient supplied by $2/x$ and $r_s x^{2s}$, $s \ge 2$, is

$$2\{(2s)(2s-1) + (2s-1)(2s-2)\} r_s = 4(2s-1)^2 r_s.$$

For $s = 1$, the first differentiation term is removed by $R''(0)$, while the second is the second derivative of a linear monomial and vanishes. This explains both the superdiagonal d_{r+1} and the exceptional indicator in (8.16).

Suppose first that $M_{q-2,q} v = 0$. The first b odd coefficients of $\mathcal{L}_Q$ vanish, hence $Y_1 = \cdots = Y_b = 0$. The first $b-1$ equations in (8.15) force

$$r_s = \rho_s r_1, \qquad 1 \le s \le b.$$

The last equation, using $r_{b+1} = 0$, then gives

$$0 = -\sum_{s=1}^{b} w_{bs} \rho_s r_1 = -d_{b+1} \rho_{b+1} r_1.$$

All factors preceding r_1 are nonzero. Therefore $r_1 = 0$, then $R = Q = 0$ and $v = 0$. This already proves nonsingularity over $\mathbb{C}$, and therefore over $\mathbb{Q}$.

To retain the exact determinant, let H_b be the matrix of (8.15) from $(r_1, \dots, r_b)$ to $(Y_1, \dots, Y_b)$. Replace its first column basis vector by $(\rho_1, \dots, \rho_b)^{\mathsf{T}}$, leaving the others fixed. The basis change has determinant one. Its first column image has only the final entry $-d_{b+1}\rho_{b+1}$. The first $b-1$ rows on columns $2, \dots, b$ form a lower triangular matrix with diagonal $d_2, \dots, d_b$. Expansion along the first column yields

$$\det H_b = (-1)^b \rho_{b+1} \prod_{s=2}^{b+1} d_s = (-1)^b 4^b \big((2b+1)!!\big)^2 \rho_{b+1}. \tag{8.17}$$

On the other hand, coefficient comparison in (8.13) and (8.12) gives

$$Y_r = \frac{2(-1)^{r-1}}{(2r-1)!} (M_{q-2,q} v)_r, \qquad r_s = \frac{(-1)^s}{(2s)!} v_{b-s}. \tag{8.18}$$

The reversal of the b columns has determinant $(-1)^{b(b-1)/2}$. Combining the diagonal factors in (8.18) gives

$$\det M_{q-2,q} = \frac{(-1)^{b(b+1)/2}}{4^b b!} \det H_b.$$

Substitute (8.17); the resulting sign is $(-1)^{b(b-1)/2}$, proving (8.8). The block decomposition and triangular transform preceding the theorem prove (8.9) and (8.10).

8.4 Explicit reconstruction without matrix inversion

The same positive recurrence supplies an exact reconstruction algorithm. Given $C_b, \dots, C_{2b-1}$, define

$$Y_r = \frac{2(-1)^{r-1}}{(2r-1)!} C_{b+r-1}, \qquad 1 \le r \le b.$$

Put $\eta_1 = 0$ and compute, for $1 \le r < b$,

$$\eta_{r+1} = \frac{Y_r + \sum_{s=1}^{r} w_{rs}\eta_s}{d_{r+1}}. \tag{8.19}$$

Then

$$\begin{aligned} r_1 &= -\frac{Y_b + \sum_{s=1}^{b} w_{bs}\eta_s}{d_{b+1}\rho_{b+1}}, \\ r_s &= \rho_s r_1 + \eta_s \quad (1 \le s \le b), \\ Z_k &= (-1)^{b-k}(2b-2k)!\, r_{b-k} \quad (0 \le k < b). \end{aligned} \tag{8.20}$$

The equations follow by writing the solution of (8.15) as $r_s = \rho_s r_1 + \eta_s$, then using the final row. The denominator in (8.20) is strictly positive by the theorem. Equations (8.2) and (8.20) express every Z_k as an explicit rational linear combination of the first q moments. Substitution into (8.3) and (8.1) recovers every later moment.

For illustration, the first five late-block determinants are

(p,q)	$(2,4)$	$(4,6)$	$(6,8)$	$(8,10)$	$(10,12)$
$\det M_{p,q}$	$-11/48$	$-113/384$	$2509/5120$	$309713/307200$	$-7097067/2867200.$

Their nontrivial numerators now have an exact origin in the positive recurrence (8.7). Finite rational checks verify these examples, the general determinant formula through the supplied test range, the rotation factors, and the reconstruction algorithm. The infinite proof is the coefficient argument above, not the finite check.

8.5 An infinite family of ordinary rational evaluations

The reconstruction has an arithmetic consequence at every adjacent pair, not just at the example printed below. Let $q = 2b$, $p = q - 2$, and denote the first row of $M_{p,q}^{-1}$ by $(\lambda_b, \dots, \lambda_{q-1})$. These rational coefficients are explicitly computed by (8.19)–(8.20). Define

$$w_r = \sum_{j=\max\{b,r\}}^{q-1} \lambda_j \binom{2j+1}{2r} 2^{-2(j-r)}, \qquad P_b(x) = \sum_{r=0}^{q-1} w_r x^{2r}. \tag{8.21}$$

Put

$$u_0 = \frac{1}{(p-1)(q-1)}, \qquad u_1 = -\frac{1}{24}\left(\frac{q}{p-1} + \frac{p}{q-1}\right).$$

Corollary 8.2 (Rational evaluations from every adjacent pair)**.** *For every even $q \ge 4$, with $p = q - 2$ and $x_n = n + 1/2$,*

$$\begin{aligned} \sum_{n=0}^{\infty} \Bigg\{ &\frac{P_b(x_n)}{(p-1)!(q-1)!}\psi^{(p-1)}(n+1)\psi^{(q-1)}(n+1) \\ &- u_0 w_{q-1} x_n^2 - u_0 w_{q-2} - u_1 w_{q-1} \Bigg\} = -\frac{1}{2}. \end{aligned} \tag{8.22}$$

The series is absolutely convergent. Its polynomial has exact degree $2q-2$, since

$$w_{q-1} = \frac{b(4b-1)\rho_b}{(2b+1)^2\rho_{b+1}} > 0. \tag{8.23}$$

Proof. By the defining inverse row, $\sum_{j=b}^{q-1}\lambda_j C_j = Z_0 = -1/2$. The inverse Bernoulli transform (8.2) then gives $\sum_{r=0}^{q-1} w_r D_r = -1/2$. For centered $x = n + 1/2$, the two leading terms of the tail product are

$$A_p(n)A_q(n) = u_0 x^{2-p-q} + u_1 x^{-p-q} + O(x^{-p-q-2}).$$

Because $p + q = 2q - 2$, multiplication by P_b leaves precisely the quadratic and constant terms subtracted in (8.22). The remainder is $O(x^{-2})$. Both subtracted terms have continued centered sum zero, so the convergent series equals the specified moment combination. The polygamma normalization follows from the two even orders. Finally the coefficient of C_{q-1} in the reconstruction of Z_0 is $b\rho_b/((2b+1)^2\rho_{b+1})$, as follows directly from the last row of (8.20). Multiplication by $\binom{2q-1}{2q-2} = 2q - 1 = 4b - 1$ proves (8.23). □

8.6 A convergent rational polygamma identity at orders four and six

The first nontrivial new adjacent example after $(2, 4)$ is $(4, 6)$. In this case (8.4) and the exact reconstruction give

$$\begin{pmatrix} C_3 \\ C_4 \\ C_5 \end{pmatrix} = \begin{pmatrix} 0 & 3/4 & 1/4 \\ 5/12 & 2/3 & -1/12 \\ 1 & -3/8 & 1/12 \end{pmatrix} \begin{pmatrix} -1/2 \\ \zeta(2) \\ \zeta(4) \end{pmatrix}, \tag{8.24}$$

and the inverse matrix is

$$\frac{1}{339}\begin{pmatrix} -28 & 180 & 264 \\ 136 & 288 & -120 \\ 948 & -864 & 360 \end{pmatrix}.$$

The first row, followed by (8.2), yields

$$67D_0 + 15516D_1 + 324320D_2 + 1434496D_3 + 1601280D_4 + 371712D_5 = -21696. \tag{8.25}$$

This has an ordinary, absolutely convergent realization. Put

$$P(x) = 67 + 15516x^2 + 324320x^4 + 1434496x^6 + 1601280x^8 + 371712x^{10}, \qquad x_n = n + \tfrac{1}{2}.$$

Then

$$\boxed{\sum_{n=0}^{\infty}\left\{\frac{P(x_n)}{720}\psi^{(3)}(n+1)\psi^{(5)}(n+1) - \frac{123904}{5}x_n^2 - \frac{316928}{5}\right\} = -21696.} \tag{8.26}$$

To check the subtraction and the meaning of this identity, the centered Hurwitz expansion gives

$$A_4(n)A_6(n) = \frac{1}{15}x_n^{-8} - \frac{7}{60}x_n^{-10} + O(x_n^{-12}).$$

Multiplication by P leaves precisely the two nonnegative powers displayed in (8.26); the remaining bracket is $O(x_n^{-2})$. Their continued centered sums are zero because $\zeta(-2, 1/2) = \zeta(0, 1/2) = 0$. Thus the absolutely convergent sum equals the meromorphically continued moment combination in (8.25). Finally $A_4(n) = \psi^{(3)}(n+1)/6$ and $A_6(n) = \psi^{(5)}(n+1)/120$, explaining the factor 720. Every summand is an ordinary real number; no finite-part instruction remains in (8.26).

There is also a finite telescoping proof of (8.26) which is independent of the continued-moment formulas. Define

$$\begin{aligned} K(N) &= -3584N^7 + 23040N^9 + 33792N^{11}, \\ P_4(N) &= 128N(N+1)(44N^4 + 88N^3 + 67N^2 + 23N - 14), \\ P_6(N) &= 128N^3(N+1)^3(33N^2 + 33N - 14), \\ R_0(N) &= -\frac{74624}{15}N^3 + 4928N^2 - \frac{244288}{15}N. \end{aligned}$$

Put

$$\mathcal{F}(N) = K(N)A_4(N)A_6(N) + P_4(N)A_4(N) + P_6(N)A_6(N) + R_0(N).$$

The identities $A_p(N-1) = A_p(N) + N^{-p}$, together with direct polynomial multiplication, give

$$\mathcal{F}(N) - \mathcal{F}(N-1) = P(N-\tfrac{1}{2})A_4(N-1)A_6(N-1) - \frac{123904}{5}(N-\tfrac{1}{2})^2 - \frac{316928}{5}. \quad (8.27)$$

This equality is a rational-function identity in N and in two indeterminates replacing $A_4(N)$ and $A_6(N)$; no zeta relation is used to verify it. Since $\mathcal{F}(0) = 0$, the partial sum of (8.26) through $n = N-1$ is exactly $\mathcal{F}(N)$. The standard uncentered Hurwitz expansions

$$A_4(N) = \frac{1}{3N^3} - \frac{1}{2N^4} + \frac{1}{3N^5} + O(N^{-7}),$$
$$A_6(N) = \frac{1}{5N^5} - \frac{1}{2N^6} + \frac{1}{2N^7} + O(N^{-9})$$

give $\mathcal{F}(N) = -21696 + O(N^{-1})$. All positive powers cancel; the constant contributions from KA_4A_6, P_4A_4, and P_6A_6 are respectively -20224, -448, and -1024. This proves the evaluation by an independent finite identity followed by an elementary asymptotic limit.

Scope. This proves the source conjecture for two infinite classes: the source's diagonal class and the new adjacent-order class. The phrase "optimal prefix" refers to the formal coordinate presentation; it does not prove that q actual numerical constants are independent. For larger gaps the rotated equation has several free initial coefficients. Positivity of its scalar recurrence alone does not establish nonsingularity of the resulting multi-column boundary matrix. That remaining distinction is essential and prevents an unsupported extension of the present theorem to all unequal pairs.

8.7 A smaller boundary determinant for a fixed gap

The rotation argument has a useful exact extension even though it does not settle the complete unequal-order conjecture. Write $q = 2b$, $p = 2(b-c)$, with $1 \le c < b$, so the differential order in the source operator is $m = 2c$. For integers $s \ge c+1$ put

$$A_s = \frac{(2s)!}{(2s-2c)!}, \qquad B_s = \frac{(2s-1)!}{(2s-2c-1)!}, \qquad d_s = 2(A_s + B_s),$$

and put $A_s = 0$ for $1 \le s \le c$. Define c fundamental rational sequences by

$$R_{s,j} = \delta_{sj}, \qquad 1 \le s, j \le c,$$
$$R_{s,j} = \frac{1}{d_s}\sum_{u=1}^{s-1} \alpha_{s-u}(B_s + A_u)R_{u,j}, \qquad s \ge c+1, \quad 1 \le j \le c. \quad (8.28)$$

Every generated entry $R_{s,j}$ is strictly positive. The relevant boundary matrix has size c, independent of b:

$$F_{b,c} = (R_{b+i,j})_{1 \le i,j \le c}.$$

Proposition 8.3 (Fixed-gap boundary reduction)**.** *In the normalization* (8.5)*,*

$$\det M_{2(b-c),2b} = \frac{(-1)^{b(b-1)/2}}{4^b b!}\left(\prod_{s=c+1}^{b+c} d_s\right)\det F_{b,c}. \quad (8.29)$$

In particular the optimal-prefix conjecture for a given gap $2c$ *is equivalent to* $\det F_{b,c} \neq 0$ *for every integer* $b > c$*. When* $c = 1$*,* $R_{s,1} = \rho_s$ *and this is Theorem 8.1.*

Proof. For $R(x) = Q(ix) = \sum r_s x^{2s}$, rotate the source operator to

$$\left(R^{(2c)} - R^{(2c)}(0)\right)\cot(x/2) + \left(R\cot(x/2)\right)^{(2c)}.$$

The coefficient of x^{2k-1}, with $s = k + c$, is

$$d_s r_s - \sum_{u=1}^{s-1} \alpha_{s-u}(B_s + A_u) r_u. \tag{8.30}$$

The pole $2/x$ gives d_s; every other cotangent coefficient has the negative sign in (8.14). The constant derivative subtraction again makes $A_c = 0$, and all lower derivatives already vanish. This proves the displayed coefficient formula.

The first $b - c$ vanishing rows successively determine $r_{c+1}, \dots, r_b$ from $r_1, \dots, r_c$ by (8.28). The remaining c rows, with $r_{b+1} = \cdots = r_{b+c} = 0$, are equivalent to the vanishing of the boundary coordinates $F_{b,c}(r_1, \dots, r_c)^{\mathsf{T}}$. For the determinant, the first elimination contributes $\prod_{s=c+1}^{b} d_s$. The remaining boundary block differs from $-\operatorname{diag}(d_{b+1}, \dots, d_{b+c})F_{b,c}$ by a unit lower triangular row transformation; its determinant is $(-1)^c \prod_{s=b+1}^{b+c} d_s \det F_{b,c}$. Moving the c initial columns past the other $b - c$ columns contributes $(-1)^{c(b-c)}$.

Finally, the rotation gives the row factor $2(-1)^{r-c}/(2r-1)!$ and the same reversed column factors $(-1)^s/(2s)!$ as before. The resulting conversion from the coefficient matrix in (8.30) to M has factor $(-1)^{b(b-1)/2+bc}/(4^b b!)$. All additional signs multiply to $(-1)^{c(1-c)} = 1$, proving (8.29). □

The positivity of all individual entries in (8.28) does not prove the determinant nonzero. There is not even a uniform positive determinant sign for every fixed gap: exact evaluation at $c = 3$ gives $\det F_{6,3} < 0$ and $\det F_{7,3} > 0$. Both are nonzero. The remaining research question is therefore a genuine boundary determinant problem, rather than a direct extension of the one-dimensional positivity proof.

9 Source audit, proof status, and further research

9.1 What has changed relative to the source questions

The baseline for this continuation is the repository revision

`f7e35be357654198c1546fd40f363694c5bcc2cc`,

dated 11 October 2026. The two paths named in the research request were inspected at that revision. Six current incoming archives were extracted for comparison, and relevant canonical chapters, the manuscript overview, and its proof-status records were consulted. The source manifest in the package records their paths and hashes. This fixes what *existing* means in the comparison; it does not imply that all historical drafts were audited.

The main advancements have different logical forms and should be integrated accordingly.

1. The higher-power request for differential-polynomial finite parts receives an all-polynomial theorem, Theorems 2.1, 2.2, and 2.3. The new scalar projection calculation identifies exactly which polynomial part produces the nondecaying residue and its Stieltjes correction. Mixed monomials have a finite, individually convergent word reduction by Theorem 3.1. The scalar power case is recovered prior work.

2. The canonical report's question about integration by parts at all weights has the exact direct-sum solution in Theorem 4.1. The endpoint map is retained in (4.6). This solves the formal quotient problem; classification of the numerical residue-coordinate image still requires relations among the resulting convergent spectral derivatives.

3. The scalar Gamma–Appell question is sharpened and partly settled at every degree by Theorem 6.1: the highest height-one Laurent coefficient is the only *singular* coordinate needed. The explicit sixth and seventh formulas implement the requested higher-degree calculations. Elimination of all convergent mixed coordinates is a different assertion, and remains open.

4. The common-shift problem receives normalized primitives and a proved endpoint limit in Theorem 5.1. This establishes the precise subtraction and its limiting constant. A reduction of every shifted primitive to a chosen multiple-Hurwitz or Barnes basis is not proved here.

5. Q8 of the continuous-depth report receives the requested all-order Gamma–Stieltjes–Z_ρ-jet answer in Theorem 7.1. The reflected jets are supplied with their own convergent evaluator in Theorem 7.2, and the result extends to ordinary Gamma-weighted polylogarithm derivatives.

6. Conjecture `conj:firstq` of the canonical fifth-moment report is proved for all $p = q - 2$ by Theorem 8.1. The general unequal-order conjecture is reduced to an explicit fixed-gap boundary determinant by Proposition 8.3, but its nonvanishing is not proved for all larger gaps. The rational polygamma-product evaluations in Section 8 include an independently telescoping example.

9.2 Corrections and editorial recommendations

No newly identified false theorem is asserted for the inspected portions of the six incoming reports. The existing cautions about finite parts, formal rank, and differentiation of discrete identities are sound. The following changes are proposed for their integration and for the research-question lists.

First, update the optimal-prefix conjecture to record the proved neighboring-order family while retaining the arbitrary-gap statement as a conjecture. Its determinant and the exact binomial omission convention should be copied together: retaining the lower-index-zero summand in (8.4) changes the matrix and invalidates the proof. Do not replace formal rank by independence of the actual tail moments or of the even zeta values.

Second, state integration-by-parts reductions as $\mathcal{I}(P) = E(Q) + \mathcal{I}(N)$, with Q at the prescribed weight. For example, $\mathcal{I}(u_1^2 + u_0 u_2) = -\zeta(3)$, not zero. This is an explicit counterexample to discarding the endpoint constant; the source question already correctly requires that constant. The new normal-form theorem provides the requested implementation of that requirement.

Third, descriptions of Gamma–Appell cancellation should distinguish the singular coordinates from convergent ones. The formulas (6.16) and (6.18) contain genuine free-order derivatives of admissible multiple zeta functions. Neither their appearance nor a failed numerical relation search proves that they are irreducible. Conversely, calling the singular support a singleton does not state that the whole answer contains only one nonclassical constant.

Fourth, add the local identity (7.11) next to the source's odd-lattice formula before discussing order derivatives. The known polynomial values in the discrete index determine the $q = 0$ case only. The reflected derivatives $\partial_s^j Z_\rho(-2m)$ are required for $q > 0$. The same care applies to the normalization factor 2^{1-s} in (7.27): its derivatives produce the explicit $\log 2$ terms in the Stieltjes examples.

Finally, retain the manuscript's existing distinction between its rejected old Gaussian S_8 vector and its revised S_8 candidate. The Gaussian S_6 and revised S_8 remain conjectural [1]. The symbols $\mathsf{S}_6, \mathsf{S}_7$ in this article are Gamma–Appell finite-part moments, unrelated to those Gaussian ladder labels. None of the present results changes the Gaussian proof status.

9.3 Verification and the role of the accompanying artifacts

The proofs in the article establish the all-order statements. The package also supplies exact finite algebra and independent numerical diagnostics. Their distinct purposes are recorded in machine-readable result files.

The adjacent-tail suite checks the source binomial matrix against the rotated cotangent operator, the positive recurrence against exact determinants, the full moment reconstruction, the fixed-gap factorization, and the explicit telescoping certificate. It records 2604 exact rational assertions and 90 independent symbolic cotangent-derivative assertions, with adjacent orders through $q = 36$. A deliberately altered coefficient is rejected. Finite examples do not prove the all-order determinant statement; the positive recurrence does.

The Gamma–Appell suite constructs finite Laurent data, the tail defects, and the strict word maps. Its weight-six certificate contains 49 shuffle, stuffle, and duality equations on 16 admissible words, together with an invertible rational subsystem. The separate replay verifies the certificate and the free-order directional reductions. Integer-index duality is used for ordinary values only. An additional independent calculation using ordered set partitions agrees with both printed directional formulas.

The Lerch suite compares exact Bell coefficients with direct logarithmic series, checks finite Euler values and trivial zeros, and compares negative-order Euler sums with positive-order Gamma-moment integrals. It records 48 exact cumulant checks, 45 finite Euler evaluations (including 20 odd trivial zeros), two differentiated-product checks, 30 independent mixed-jet numerical comparisons, and direct Stieltjes derivative checks. The latter comparisons include noninteger and complex depth. Their largest recorded relative residual is approximately 1.17×10^{-71}.

The differential-polynomial suite verifies all 270 formal monomial decompositions of weights 2–12, with 22 two-sided exact matrix inverse identities. It checks 128 mixed and 10 scalar Laurent residues through weight ten; 48 bilinear pairs, including every applicable zero-order case; 169 finite harmonic identities; and explicit endpoint constants. Its numerical companion computes finite parts using a convergent Taylor integral near zero and ordinary quadrature away from zero. It compares two splitting points, tests nonlinear endpoint identities, checks normalized primitives at real and complex arguments, and differentiates ordinary common-shift integrals.

All numerical records use arbitrary-precision floating-point arithmetic; they are not interval certificates. The article does not claim proof-assistant formalization. The symbolic certificates are exact finite objects, but the analytic continuation, convergence, and normalization lemmas remain mathematical proofs in the text.

9.4 Further research questions

The following questions seek identities, finite classifications, and functional relations. Estimates would serve to justify operations or certificates, rather than being the principal research target.

9.4.1 The boundary determinant for arbitrary even gaps

For each fixed $c \geq 2$, prove or disprove $\det F_{b,c} \neq 0$ for every $b > c$, with the explicit fundamental sequences in (8.28). The case $c = 2$ is the first unresolved family. The reduction is useful because the determinant size depends on the gap, not the upper tail order. A successful proof could come from a discrete Wronskian identity, a recurrence for the minors, or a factorization in a suitable polynomial basis. A uniform positivity argument for all gaps is unavailable: the sign change already recorded at $c = 3$ rules out that simplest claim.

9.4.2 Systematic rational telescoping for tail products

The explicit $(4, 6)$ identity has a certificate involving only a polynomial multiple of $A_4(N)A_6(N)$, two single-tail terms, and a polynomial. Determine whether every adjacent-order reconstruction identity admits a certificate of this form, and derive its coefficients directly from the recurrence for ρ_s. Such a theorem would replace spectral continuation by a finite telescoping proof for the whole ordinary-series family. A useful first output is a bound on the required polynomial degrees proved from the shift recurrence for the tails.

9.4.3 The spectral image of the canonical differential basis

Theorem 4.1 classifies differential monomials before arithmetic relations are imposed. Apply the convergent-word map to this basis and determine its kernel modulo a precisely stated family of free-order stuffle relations and differential identities. The first interesting weights are those containing several nonlinear mixed basis elements, such as $u_0^2u_1^2$ and u_1^3 at weight six. The desired result is a finite presentation with a proved rank in that presentation. Arithmetic independence of its evaluated numbers should remain a separate question.

9.4.4 Relations among the sixth and seventh mixed directions

Determine which combinations in (6.16) and (6.18) reduce to lower depth, ordinary zeta derivatives, or named height-one Laurent coefficients. Symmetric free-order stuffle relations already explain part of the reduction; the surviving antisymmetric directions require additional identities. Tornheim functional relations, parameter-dependent iterated integrals, and deformation of known double-zeta identities are concrete candidates. An identity known only at integer orders cannot be differentiated for this purpose.

9.4.5 Higher spectral derivatives of the residue calculus

The present finite-part map selects the coefficient of s in a Dirichlet germ. Introduce logarithmically weighted spectral functionals whose ordinary representatives select $[s^q]$ for $q \geq 2$. Determine whether the same scalar projection controls all their Stieltjes corrections, and identify a canonical family of normalized spatial kernels realizing them. For mixed monomials the strict upper–middle–lower construction already gives convergent higher partial derivatives; the remaining issue is the exact spatial normalization and its endpoint constants.

9.4.6 Rational shifts and multiple-Hurwitz coordinates

For $I_P(a)$ and $\mathcal{P}_P(a)$ at rational $a > 0$, find finite formulas in a specified algebra of multiple-Hurwitz derivatives and cyclotomic polylogarithms. The normalized kernels in (5.7) and the endpoint law (5.4) fix the constants that such a formula must respect. Reflection and multiplication combinations may simplify before individual values do. A complete result should also explain the relation to the Gamma-quotient normalization with a general base point.

9.4.7 Additional reductions at rational noninteger depth

For fixed rational noninteger $\rho > 0$, investigate the reflected coordinates $\partial_\rho^r \partial_s^q Z_\rho(-2m)$. Theorem 7.2 already evaluates them by normally convergent series of finite harmonic-logarithmic expressions. The further arithmetic question is whether selected depths admit finite expressions in a predeclared collection of Hurwitz, Gamma, or cyclotomic constants. Integer-depth termination at $q = r = 0$ does not persist automatically after differentiation. Identifying exactly when

an additional finite reduction exists would be a substantive strengthening of the current lattice theorem.

9.4.8 Independent twists and noncentered Gamma weights

Extend the local lattice construction to the noncentered and tilted families while keeping order, shift, and depth as independent complex variables. The centered theorem uses the even Taylor structure and a single sine factor. Away from the center, the reflection system can couple two branches or two transforms. Derive the corresponding local matrix identity and determine its removable values before extracting mixed jets. The desired output is a finite formula for ordinary Gamma-weighted root-of-unity polylogarithm derivatives, with every branch and normalization specified.

9.4.9 Four or more harmonic tails

For products of an even number of even-order tails, develop a finite coordinate system and prove an initial-moment generation theorem. Centered parity still controls which spectral values can be singular, but the two-tail rank calculation does not establish the higher-product claim. The relevant differential operators should contain products of cotangent kernels and higher pole multiplicities. Exact finite reconstruction and ordinary telescoping identities would be natural outputs. Odd numbers of tails require a separate finite-part normalization because their centered parity changes.

9.4.10 Functional certificates for the remaining ladders

Continue the exact functional-equation search for the manuscript's Gaussian S_6 and revised S_8 candidates, keeping the evaluated identity distinct from the formal symbol calculation. The new convergent spectral coordinates suggest additional Mellin and polylogarithmic representations, but no bridge to those particular ladder vectors is proved here. A successful next step should supply an explicit functional certificate, a proved obstruction in a stated relation system, or a rigorously nonzero evaluated residual.

9.4.11 Formal certificates with a small analytic interface

The finite algorithms admit a useful separation: exact Laurent extraction and word multiplication, rational matrix reconstruction, and the analytic lemmas for normal convergence and finite parts. Formalize the first layer independently, then connect it through proved interfaces to the scalar residue filter, the normalized Mittag–Leffler theorem, and the Euler holomorphy theorem. The purpose is a reusable proof pipeline for further identities, with each numerical diagnostic remaining a separately identified experiment.

References

[1] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*, canonical manuscript and editorial records, revision `f7e35be357654198c1546fd40f363694c5bcc2cc`, 11 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/f7e35be357654198c1546fd40f363694c5bcc2cc/Analysis/Polylogarithms/docs/manuscript.

[2] ProveIt research continuation, *Higher Digamma Powers and Ordered Gamma Sums*, 11 October 2026. Incoming archive at the same revision; especially the all-power Mittag–Leffler theorem, the finite harmonic-word algorithm, and the differential-polynomial, common-shift, and higher-moment

questions. https://github.com/VladimirReshetnikov/ProveIt/blob/f7e35be357654198c1546fd40f363694c5bcc2cc/docs/incoming/ProveIt_Higher_Digamma_Powers_2026-10-11.zip.

[3] ProveIt research continuation, *Canonical Digamma Polynomials and the Fifth Harmonic Bridge: Laurent Coefficients, Polygamma Primitives, and Finite Reconstruction of Even-Tail Moments*, 11 October 2026. Incoming archive at the same revision; Sections 4, 7, and 9, including Conjecture `conj:firstq`. https://github.com/VladimirReshetnikov/ProveIt/blob/f7e35be357654198c1546fd40f363694c5bcc2cc/docs/incoming/ProveIt_Canonical_Digamma_Fifth_Moment_2026-10-11.zip.

[4] ProveIt research continuation, *Mixed Resolvent Products, Higher Digamma Powers, and Stieltjes Collisions: Exact Identities and Normalized Finite Parts*, 11 October 2026. Incoming archive at the same revision. https://github.com/VladimirReshetnikov/ProveIt/blob/f7e35be357654198c1546fd40f363694c5bcc2cc/docs/incoming/ProveIt_Mixed_Resolvents_Harmonic_Powers_2026-10-11.zip.

[5] ProveIt research continuation, *Continuous-Depth Lerch Calculus: Gamma Kernels, All-Order Harmonic Identities, Stieltjes Convolutions, and Anchored Primitives*, 10 October 2026. Incoming archive at the same revision; the centered reflection theorem, Gamma convolution, and final research question Q8. https://github.com/VladimirReshetnikov/ProveIt/blob/f7e35be357654198c1546fd40f363694c5bcc2cc/docs/incoming/ProveIt_Continuous_Depth_Lerch_2026-10-10.zip.

[6] ProveIt research continuation, *Tilted Lerch Convolutions and Binary Harmonic Cancellation: Confluent Zeta Identities, Independent Spectral Jets, Gamma Primitives, and Mixed-Parity Exact Evaluations*, 10 October 2026. Incoming archive at the same revision. https://github.com/VladimirReshetnikov/ProveIt/blob/f7e35be357654198c1546fd40f363694c5bcc2cc/docs/incoming/ProveIt_Tilted_Lerch_Binary_Cancellation_2026-10-10.zip.

[7] ProveIt research continuation, *Algebraic Gauss Residues and Centered Harmonic Reflection*, incoming archive at the same revision. https://github.com/VladimirReshetnikov/ProveIt/blob/f7e35be357654198c1546fd40f363694c5bcc2cc/docs/incoming/Algebraic_Gauss_Centered_Harmonic_Reflection.zip.

[8] NIST Digital Library of Mathematical Functions, Chapter 5, especially §§5.7 and 5.15, Gamma expansions, polygamma recurrences, and special values. https://dlmf.nist.gov/5.7; https://dlmf.nist.gov/5.15.

[9] NIST Digital Library of Mathematical Functions, §25.11, Hurwitz zeta function. https://dlmf.nist.gov/25.11.

[10] NIST Digital Library of Mathematical Functions, §25.14, Lerch's transcendent. https://dlmf.nist.gov/25.14.

[11] NIST Digital Library of Mathematical Functions, §3.9(ii), Euler's transformation of series. https://dlmf.nist.gov/3.9#ii.

[12] K. Ihara, M. Kaneko, and D. Zagier, Derivation and double shuffle relations for multiple zeta values, *Compositio Mathematica* **142** (2006), 307–338. https://doi.org/10.1112/S0010437X0500182X. Author-hosted full text: https://www2.math.kyushu-u.ac.jp/~mkaneko/papers/34Derivation_double_shuffle.pdf.

[13] D. Chakrabarti and G. K. Srinivasan, *On a remarkable formula of Ramanujan*, 2012. https://arxiv.org/abs/1206.5122. The Gamma-product Fourier normalization used here is also derived directly from the beta integral in Section 7.